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ParkViewRT: Vision-Based Parking Slot Monitoring System

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BY

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Abstract

The problem of parking congestion is a widespread problem in university campuses, which is prone to time wastage, frustration and poor circulation of traffic. In Multimedia University (MMU), lack of clear and timely information regarding the availability of parking space often creates a problem to users in finding parking space, especially during peak hours. The proposed project will focus on solving this issue and suggest a camera-based parking monitoring system, which uses computer vision to determine the occupancy of parking slots. Survey and an observational study were to be performed to identify the parking issues at the university level by targeting high-demand parking zones, i.e., the Faculty of Computing and Informatics (FCI) and the Faculty of Artificial Intelligence and Engineering (FAIE). These analyses were applied to determine the user requirements and parking usage patterns based on which the system design was carried out. Through the analysis, a parking monitoring system was developed that works with the videos that have been recorded in order to process the video material with the help of the vehicle identification method based on YOLO and define whether a parking slot is occupied or not. The system offered has the predetermined mapping of parking slots and dashboard to display the parking plans, slot status, and overview data of the chosen parking areas. The implementation and performance evaluation of the systems will be done in the following stage of the project due to time and resource limitations. On the whole, the given work forms a systematic basis of the parking monitoring system development at MMU because it analyzes the problem, designs the system and offers the implementation plan to facilitate the future system development and evaluation.

Table of Contents

Copyright	iii
Declaration	iv
Acknowledgements	v
Abstract	vi
Table of Contents.....	vii
List of Figures	xi
List of Appendices	xiii
Chapter 1: Introduction	1
1.1 Overview	1
1.2 Problem Statement	3
1.3 Project Objectives	6
1.4 Project Scope	7
1.5 Project Limitations	8
1.6 Methodology	9
1.7 Target Audience	10
1.8 Summary	11
Chapter 2: Literature Review	13
2.1 Overview	13
2.2 Reliance on Manual Labeling and Scalability Limitations (Automatic Parking Slot Detection Gap)	20
2.3 Environmental and Illumination Challenges	23
2.4 Computational Resource Constraints and Meeting Real-Time Speed	28
2.5 False Detections from Transient or Illegal Vehicles	32
2.6 Occlusion and Generalization Across Diverse Viewpoints	35
2.7 Handling Varied Geometries and Angular Orientation	39
2.8 Advanced Classification for Suitability and Constraints	42
2.9 Dataset Limitations	45
2.10 Summary	49
Chapter 3: Requirements Analysis	53
3.1 Overview	53

3.2 Fact-Finding Techniques	53
3.2.1 Justification.....	54
3.2.2 Questionnaire Design.....	55
3.2.2.1 Analysis on results	57
a. Question 1: Role at MMU.....	58
b. Question 2: Faculty.....	58
c. Question 3: Vehicle Ownership	59
d. Question 4: Parking Frequency	60
e. Question 5: Time of Highest Parking Difficulty	61
f. Question 6: Most Frequently Used Parking Area	62
g. Q7: Area with Most Difficulty Finding Parking	63
h. Q8: Main Parking Issues (Multi-select)	64
i. Q9: Time Spent Searching for Parking.....	66
j. Q10: Satisfaction with Current Parking	67
k. Q11: Desired Improvements to the Current Parking System	68
l. Q12: Importance of Knowing Parking Availability Before Arrival	69
m. Q13: Essential Features for an Optimal Parking System.....	70
n. Q14: Impact of Parking Availability Display on Reducing Stress	71
o. Q15: Previous Use of Parking Applications	72
p. Q16: Parking Application Features Used (Multi-select).....	72
q. Q17: Perceived Usefulness of a Vision-Based Parking System	73
r. Q18: Support for Implementing the System at MMU	74
3.2.3 Observation.....	76
a. FCI Parking Observation	77
b. FAIE Parking Observation.....	79
3.2.3.1 Analysis on results	81
3.3 Requirement	82
3.3.1 Functional Requirements.....	83
a. FR-1: Parking Slot Occupancy Detection	83
b. FR-2: Video Feed Input Handling.....	83
c. FR-3: Frame Extraction from Video Feed.....	84
d. FR-4: Parking Slot Status Classification.....	84

e.	FR-5: Real-Time Parking Availability Update	84
f.	FR-6: Visual Representation of Parking Slot Status	84
g.	FR-7: Support for Multiple Parking Locations	85
h.	FR-8: Parking Location Selection Interface	85
i.	FR-9: Display of Parking Layout	85
j.	FR-10: Summary of Parking Availability	85
3.3.2	Non-functional Requirements	86
a.	NFR-1: System Accuracy	86
b.	NFR-2: System Response Time	86
c.	NFR-3: Usability and Ease of Use	87
d.	NFR-4: Scalability	87
3.3.3	User Requirements	87
a.	UR-1: Access to Clear Parking Availability Information	87
b.	UR-2: Simple and Intuitive Interface	88
c.	UR-3: Minimal Time and Effort to Obtain Information	88
d.	UR-4: Confidence in Displayed Information	88
Chapter 4:	System Design	90
4.1	Overview	90
4.2	Rich Picture Diagram	90
4.3	Use Case Diagram	92
4.4	Activity Diagram	93
4.5	Class Diagram	94
4.6	Sequence Diagram	95
4.7	Interface Design	97
4.8	Summary	100
Chapter 5:	Implementation Plan	101
5.1	System Implementation Overview	101
5.2	Video Data Collection	102
5.3	Parking Occupancy Detection Model	103
5.4	Parking Slot Mapping and Status Update	105
5.4.1	Parking Slot Region Definition	105
5.4.2	Mapping Detection Results to Parking Slots	105

5.4.3	Parking Slot Status Update Algorithm	106
5.4.4	Status Synchronization with Dashboard	107
5.5	Dashboard Implementation	107
5.6	Tools and Technologies	108
5.6.1	Programming Language.....	108
5.6.2	Video Processing.....	109
5.6.3	Parking Occupancy Detection.....	109
5.6.4	Mapping Parking Slots and Determining Statuses.	109
5.6.5	Dashboard Visualization	110
5.7	Summary	110
5.7.1	Video Data Collection	110
5.7.2	Parking Occupancy Detection.....	111
5.7.3	Status Update and Parking Slot Mapping.....	111
5.7.4	Dashboard Implementation.....	111
5.7.5	Tools and Technologies	112
Chapter 6:	Conclusion	113
6.1	Project Summary.....	113
6.2	Achievement of Project Objectives	113
6.3	Key Outcomes and Current Limitations.....	114
6.4	Future Work and Next Phase (FYP2)	115
6.5	Concluding Remarks	116
References		xii
Appendix A:	Gantt Chart.....	16
Appendix B:	FYP I Meeting Logs.....	17
Appendix C:	Turnitin Similarity Index Page.....	38

List of Figures

Figure 3.1: Respondents Role	58
Figure 3.2: Respondents Faculties	59
Figure 3.3: Vehicle Ownership	60
Figure 3.4: Parking Frequency	61
Figure 3.5: Times of Highest Parking Difficulty	62
Figure 3.6: Most Frequently Used Parking Area	63
Figure 3.7: Distribution of areas where respondents experience the greatest difficulty in finding parking at MMU	64
Figure 3.8: Summary of the main parking issues reported by respondents.	65
Figure 3.9: Distribution of time spent by respondents searching for parking spaces at MMU	66
Figure 3.10: Respondents' satisfaction levels with the current parking system at MMU	67
Figure 3.11: Respondents' desired improvements to the current MMU parking system.	68
Figure 3.12: Perceived importance of knowing parking availability among respondents	69
Figure 3.13: Essential features identified by respondents for an optimal parking system	70
Figure 3.14: Respondents' perceptions of whether parking availability displays can reduce parking-related stress.	71
Figure 3.15: Distribution of respondents based on prior usage of parking applications	72
Figure 3.16: Frequency of parking application features used by respondents..	73
Figure 3.17: Respondents perceived usefulness of a vision-based parking system.	74
Figure 3.18: Level of respondent support for implementing a smart parking system at MMU	74
Figure 3.19: FCI Parking (8 - 10 am)	77

Figure 3.20: FCI Parking (10 - 2 pm)	78
Figure 3.21: FCI Parking (2 - 5 pm)	78
Figure 3.22: FAIE Parking (8 - 10 am)	79
Figure 3.23: FAIE Parking (10 - 2 pm)	80
Figure 3.24: FAIE Parking (2 - 5 pm)	80
Figure 4.1: Rich Picture Diagram.....	91
Figure 4.2: Use Case Diagram	92
Figure 4.3: Activity Diagram.....	93
Figure 4.4: UML Class Diagram.....	95
Figure 4.5: Sequence Diagram.....	96
Figure 4.6: Home Page Interface.....	97
Figure 4.7: FCI Parking Interface.....	98
Figure 4.8: FAIE Parking Interface.....	99
Figure: Gantt Chart for FYP2.....	105
Figure: Turnitin Similairy %.....	125

List of Appendices

Appendix A: Gantt Chart	Error! Bookmark not defined.
Appendix B: FYP I Meeting Logs	Error! Bookmark not defined.
Appendix C: Turnitin Similarity Index Page	Error! Bookmark not defined.

Chapter 1: Introduction

1.1 Overview

The development of the **ParkViewRT: Vision-Based Parking Slot Monitoring System** aligns with a global shift toward **Smart City infrastructures**, where the efficient management of urban mobility is paramount. As urban populations surge, with the UN estimating that 68% of the world will reside in cities by 2050, the demand for parking has reached a crisis point. Drivers in metropolitan areas spend an average of **17 hours per year searching for parking**, a process that intensifies traffic congestion, wastes fuel, increases CO2 emissions, and elevates driver stress. Vision-based real-time monitoring represents a transformative solution to these challenges, offering a scalable and cost-effective alternative to traditional management methods.

Historically, parking systems relied on physical sensors (ultrasonic, magnetic, or infrared) installed in every individual slot. While accurate, these hardware-centric solutions are plagued by high installation and maintenance costs; for instance, the SF Park system in San Francisco cost approximately \$1,400 per sensor. Furthermore, sensors are prone to environmental damage, have limited lifespans, and cannot provide rich contextual data.

In contrast, vision-based systems leverage high-definition cameras or existing CCTV infrastructure to monitor dozens of parking spaces simultaneously. This approach is significantly more scalable and unobtrusive, providing additional functionalities such as vehicle recognition, security surveillance, and the detection of abnormal behavior or parking violations.

The literature generally categorizes the tasks of a vision-based parking system into three primary areas:

- **Automatic Parking Space Detection (Localization):** The ability to autonomously identify the coordinates and boundaries of parking slots without manual labeling, which is crucial for system scalability.

- **Occupancy Classification:** The binary task of determining whether a detected slot is vacant or occupied by a vehicle.
- **Vehicle Counting and Suitability Assessment:** Advanced systems now strive to count total vehicles in a lot and classify slots based on mission-specific constraints, such as **handicap accessibility or EV charging** availability.

Developing a reliable real-time system like **ParkViewRT** requires overcoming several "open-air" challenges:

- **Environmental Variability:** Systems must maintain high accuracy across diverse lighting conditions (day vs. night) and severe weather patterns (rain, snow, fog, or haze).
- **Physical Occlusions:** Vision can be obstructed by stationary objects like trees and lampposts, or dynamic factors such as adjacent parked vehicles and transient traffic.
- **Geometric Diversity:** Parking layouts are rarely uniform, featuring perpendicular, parallel, and **slanted (fishbone)** geometries that require sophisticated polygon-based representations rather than simple bounding boxes.
- **Perspective Distortion:** Cameras mounted at various heights and angles introduce lens distortion and perspective shifts, necessitating the use of **Inverse Perspective Mapping (IPM)** to generate a normalized bird's-eye view for analysis.

The research has transitioned from traditional image processing (e.g., Canny edge detection and Haar features) to **Deep Learning (DL)** frameworks. Currently, the **YOLO (You Only Look Once)** family of models (v5 through v11) dominates the field due to its ability to perform object detection and classification

in a single forward pass, ensuring the high-speed inference required for real-time monitoring.

To ensure practical deployment, modern research emphasizes **Edge Intelligence**. This involves deploying lightweight models (such as mAlexNet, YOLO-tiny, or optimized MobileNetV3) directly onto embedded devices like the **Raspberry Pi or Nvidia Jetson**. This decentralized approach reduces communication overhead, eliminates server bottlenecks, and preserves data privacy by processing images locally and only transmitting binary occupancy status to the cloud.

The **ParkViewRT** system sits at the intersection of these advancements, utilizing **real-time computer vision** and **deep learning** to provide a robust, non-invasive monitoring solution. By integrating automated spatial detection with efficient classification, the system addresses the critical need for a scalable, 24/7 parking management tool that contributes to the overarching goals of urban sustainability and optimized traffic flow.

1.2 Problem Statement

The development of the **ParkViewRT: Vision-Based Parking Slot Monitoring System** is necessitated by several critical failures in current urban parking management and technological implementation. Despite the rapid growth of urban populations and vehicle ownership, the infrastructure required to manage these vehicles remains largely inefficient and outdated. The following points determine the core components of the problem:

a) Economic and Environmental Costs of "Cruising"

In highly densely populated urban areas, the search for available parking often termed as "**cruising**," has reached a state of crisis. Motorists spend an average of **17 hours per year** searching for parking, a number that escalates to **107 hours** in cities like New York. This inefficient process results in approximately

\$345 per driver in wasted time, fuel, and excess emissions, while simultaneously intensifying urban traffic congestion and increasing the risk of accidents. Drivers in these scenarios experience heightened levels of stress and divided attention, which significantly degrades both individual well-being and general road safety.

b) Limitations of Physical Sensor Infrastructure

Traditional automated parking systems have relied heavily on **ground-based physical sensors** (e.g., ultrasonic, magnetic, or infrared) installed in every individual slot. While functionally accurate, these hardware-centric solutions are prohibitively expensive to install and maintain; for example, San Francisco's SF Park program incurred costs of approximately **\$1,400 per sensor**. Beyond the financial burden, these sensors are prone to **physical damage and environmental wear**, have limited lifespans and are incapable of providing the rich contextual information (such as vehicle recognition or anomaly detection) that vision-based systems offer.

c) The "Manual Labeling" Deployment Bottleneck

Existing computer vision-based solutions, while more scalable than sensors, suffer from a significant **deployment bottleneck**: the requirement for **manual labeling of parking slot locations**. For an operator managing hundreds of slots across multiple lots, manually annotating every coordinate is an exhaustive and inefficient expenditure of labor. Furthermore, these systems lack flexibility; if a camera's perspective shifts due to maintenance, zooming, or replacement, the entire manual labeling process must be repeated from scratch, rendering current systems **unscalable for large-scale urban deployment**.

d) Technical Fragility in Complex Environments

Real-world parking environments present a variety of "open-air" challenges that cause many current algorithms to fail. Lighting variations, heavy shadows, and adverse weather conditions (such as **rain, snow, or fog**) create textural patterns that frequently lead to misclassification. Additionally, **physical occlusions**, where trees, lampposts, or adjacent vehicles partially block the camera's view which hinders the system's ability to recognize a slot's status. Many systems also rely on rigid rectangular bounding boxes that fail to precisely capture the area of **slanted or fishbone parking slots**, which are common in modern parking lot designs.

e) Hardware Constraints and Data Privacy

Executing sophisticated deep learning models in **real-time on edge devices** (such as a Raspberry Pi or Nvidia Jetson) presents a difficult computational challenge. Large-scale models designed for server-level performance often cannot meet the low-latency requirements of a decentralized smart camera network. Furthermore, there are growing **privacy concerns** regarding the transmission of high-resolution images of citizens and vehicles to central cloud servers, creating a need for localized, on-device processing that only transmits binary occupancy data.

f) The "Suitability" Gap

Finally, the majority of research in this field focuses exclusively on the binary question of whether a spot is **occupied or vacant**. This ignores a critical aspect of intelligent vehicle navigation: determining if a vacant spot is **compatible with the vehicle's mission**. Current systems largely fail to identify and classify spots designated for specific needs, such as **handicap accessibility or electric vehicle (EV) charging** stations, which is a fundamental requirement for the realization of fully autonomous valet parking.

Relying on current parking management systems is like trying to manage a massive library by hiring a single person to manually write down the location of every single book on a sheet of paper. If the librarian moves a shelf just an inch, the entire list becomes useless and must be thoroughly rewritten. Moreover, this librarian can only tell you if a shelf has a book on it but can't tell you if that book is the specific one you actually need, or if the shelf is even strong enough to hold your particular volume.

1.3 Project Objectives

The objectives of this project are defined with consideration to the scope, time constraints, and practical limitations of a Final Year Project. The primary aim is to design and implement a prototype-level system that demonstrates the feasibility of vision-based parking slot monitoring in a real-world campus environment.

As part of this study, a limited number of parking slots within the Multimedia University (MMU) campus will be selected to evaluate the proposed approach. Video feeds will be captured using mobile or aerial perspectives at different times of the day in order to observe variations in lighting conditions and potential environmental challenges.

- To identify a small set of parking slots (approximately two to four) within the MMU campus using the survey findings as a selecting criterion.
- To collect parking area video data using mobile or drone-based recordings captured at different times of the day to reflect real-world usage conditions.
- To design and implement a computer vision–based model capable of detecting vehicle presence and determining parking slot occupancy status.
- To analyze the impact of environmental factors such as lighting variations, occlusions, visual obstacles and camera perspective on detection performance.
- To visualize the occupancy status of selected parking slots through a simple dashboard or graphical interface for monitoring purposes that mimics the selected parking slot.

1.4 Project Scope

The scope of this project is focused on the development and evaluation of a prototype vision-based parking slot monitoring system within a controlled campus environment. The implementation is limited to a small number of selected parking slots at Multimedia University (MMU) to ensure the project remains feasible within the duration and resources of a Final Year Project.

The system will utilize pre-recorded video feeds captured using mobile devices or aerial platforms such as drones. These recordings will be taken at different times of the day in order to observe variations in lighting conditions, shadows and vehicle movement patterns. The collected footage will be used to test and evaluate the performance of the proposed detection approach.

The project scope includes the design and implementation of a computer vision model to detect vehicle presence and determine the occupancy status of

predefined parking slots. The system will map detection results to specific parking slots and visually indicate whether each slot is occupied or vacant through a basic graphical interface or dashboard.

The scope of this project does not include large-scale deployment, integration with existing university parking management systems, or the use of permanent camera infrastructure. Additionally, the system is not intended to provide real-time guidance to drivers or support payment, reservation or enforcement functionalities. The focus of the project is limited to demonstrating the feasibility and effectiveness of vision-based parking slot monitoring at a prototype level.

1.5 Project Limitations

This project is developed within the constraints of a Final Year Project and is therefore subject to several limitations related to scope, data collection and system implementation. These limitations are acknowledged to ensure realistic expectations of the system's performance and applicability.

The key limitations of this project are as follows:

- The system is implemented and evaluated using a limited number of parking slots (approximately two to four) within the Multimedia University (MMU) campus, which may not represent larger or more complex parking environments.
- Video data is collected using mobile or drone-based recordings rather than fixed, permanently installed cameras, resulting in variations in camera angle, height and stability that may affect detection accuracy.
- Environmental factors such as lighting changes, shadows, weather conditions and partial occlusions may impact the reliability of vehicle detection under certain scenarios.

- The project focuses on a prototype-level implementation to demonstrate feasibility and does not aim to achieve commercial-grade performance or large-scale deployment.

In conclusion, these limitations define the boundaries of the current study and reflect practical constraints faced during project development. Despite these limitations, the project provides valuable insights into the feasibility of vision-based parking slot monitoring and establishes a foundation for future improvements and extended implementations.

1.6 Methodology

This project adopts an application-based development methodology that combines data collection, system modelling and prototype implementation to evaluate the feasibility of a vision-based parking slot monitoring system. The methodology is structured into several sequential phases to ensure a systematic and organized development process.

The first phase involves **site selection and data collection**. A small number of parking slots within the Multimedia University (MMU) campus are identified as the study area. Video data of the selected parking slots is captured using mobile devices or aerial platforms at different times of the day. This approach allows the system to be evaluated under varying lighting conditions and real-world usage scenarios.

The second phase focuses on **data preparation and analysis**. The collected video recordings are processed to extract relevant frames for model training and evaluation with addition to publicly available datasets. Parking slot regions are defined manually to establish ground truth information, which serves as a reference for occupancy detection and performance assessment.

The third phase involves **model development and system integration**. A computer vision-based detection model is designed to identify vehicle presence

within the captured video frames. The detection results are mapped to predefined parking slots to determine occupancy status. The system integrates detection outputs with a visual representation that indicates whether each parking slot is occupied or vacant.

The fourth phase consists of **system testing and evaluation**. The prototype system is tested using video data captured under different environmental conditions to evaluate detection accuracy and robustness. Observations are made regarding system performance, potential errors and environmental challenges such as lighting variations and occlusions.

Finally, the project concludes with **result analysis and documentation**. The outcomes of the system evaluation are analyzed to identify strengths, limitations, and areas for improvement. All findings, design decisions and results are documented to support further development in the subsequent phase of the Final Year Project.

1.7 Target Audience

The primary target audience of this project includes but is not limited to:

- Parking Administrators
- Facility Management Personnel
- University Staff responsible for monitoring and managing parking spaces within campus environments.
- Shopping Malls Management
- Real-estate Companies

The proposed system is designed to assist these stakeholders by providing a clear visual representation of parking slot occupancy, enabling more efficient observation and utilization of available parking spaces.

In addition, this project is relevant to researchers and students interested in computer vision–based intelligent transportation systems, particularly those

exploring vision-driven monitoring solutions in constrained environments. The prototype also serves as a reference implementation for future academic projects involving smart campus infrastructure and visual analytics.

1.8 Summary

This chapter has presented an overview of the project by introducing the problem of inefficient parking space monitoring and the growing need for intelligent, automated solutions in campus environments. The motivation behind the project was discussed, highlighting the limitations of conventional parking monitoring methods and the potential of computer vision techniques to improve parking space utilization and situational awareness.

The project objectives were clearly defined with consideration to the scope and constraints of a Final Year Project. These objectives focus on implementing and evaluating a prototype vision-based parking slot monitoring system using a limited number of parking spaces within the Multimedia University (MMU) campus. The scope of the project was subsequently outlined to clarify the boundaries of the system, ensuring that the proposed solution remains feasible while still addressing the core problem effectively. Key limitations related to data collection methods, environmental factors and deployment scale were also identified to provide a realistic understanding of the system's capabilities.

In addition, a high-level methodology was described to outline the structured phases of the project, including data collection, model development, system integration and performance evaluation. This methodology provides a systematic approach to achieving the project objectives while maintaining clarity and organization throughout the development process. The target audience of the system was also identified, emphasizing its relevance to campus parking administrators as well as students and researchers interested in intelligent transportation and computer vision-based monitoring systems with addition to the rest of the interested stakeholders.

This chapter establishes a clear foundation for the remainder of the project by defining the problem context, objectives, scope and development approach. The following chapters will be built upon this foundation by presenting detailed requirements analysis and system design, leading towards the implementation and evaluation of the proposed parking slot monitoring system.

This report is organized as follows: **Chapter 1** introduces the background of the project, problem statement, objectives, scope, limitations, methodology, and target audience. **Chapter 2** reviews related work and existing approaches in parking management and computer vision–based parking monitoring systems. **Chapter 3** presents the requirements analysis, detailing the system requirements derived from observation and analysis of real-world parking scenarios. **Chapter 4** describes the system design, including architectural components and relevant diagrams used to model the proposed solution. **Chapter 5** outlines the implementation plan for system development and testing. Finally, **Chapter 6** concludes the report by summarizing the project outcomes and discussing potential directions for future work.

Chapter 2: Literature Review

2.1 Overview

The management of urban parking resources is experiencing a fundamental shift, moving from legacy hardware-based systems toward cost-effective and highly scalable vision-based solutions leveraging advancements in Deep Learning (DL)(da Luz et al., 2024; Grbić & Koch, 2023). This evolution is necessitated by the rapid growth in urban populations and vehicle ownership, which has saturated urban road networks and space reserved for parking. Consequently, drivers often cruise for vacant parking slots, contributing to serious urban challenges including traffic congestion, increased pollution, emissions, and significant time and cost losses for drivers (Grbić & Koch, 2023).

To combat these issues, **Parking Guidance Information (PGI) systems** require accurate and real-time information about the occupancy of every parking slot (Grbić & Koch, 2023). Traditional approaches relied on installing hardware sensors, such as ultrasonic, magnetic, or combinations thereof, directly onto each parking slot. While these sensor-based systems can deliver **highly accurate information**, they impose **substantial costs** related to sensor hardware, installation, and ongoing maintenance. This high expenditure severely compromises the scalability of sensor-based systems, especially for large parking areas or future urban expansions (da Luz et al., 2024).

Vision-based solutions offer an appealing alternative, as a **single camera can monitor a wide parking area**, often repurposing existing surveillance infrastructure, leading to a significantly **lower cost and easier implementation**. Furthermore, vision systems can provide richer contextual information, such as vehicle recognition. However, the shift to vision-based PGI systems, particularly those utilizing deep learning (DL) and computer vision techniques, introduces complex **unique challenges** tied to their technical implementation and the unpredictable nature of real-world environments (Grbić & Koch, 2023).

Many of the foundational deep learning techniques for vision-based parking detection originate from models like **YOLO (You Only Look Once)**, which are favored for their speed and accuracy in object detection. Systems often convert the parking detection problem into two main tasks: identifying the precise location of parking spots (parking slot detection) and determining whether each spot is occupied or vacant (occupancy classification) (Amato et al., n.d.; Barseghyan, 2023; Popereshnyak & Chornobryvets, 2025; Wang et al., 2023).

Successful and widespread deployment hinges on mitigating these persistent implementation challenges. Despite the advantages of vision-based parking detection systems, developing and deploying robust vision-based systems introduce their own unique set of technical and practical challenges. The current literature highlights several critical and unique problems that developers and researchers must overcome to create robust and reliable vision-based parking systems:

a) Reliance on Manual Labeling and Scalability Limitations (Automatic Parking Slot Detection Gap)

A significant obstacle researchers highlight in deploying vision-based parking slot detection is the extensive requirement for manual labeling of parking slot locations in input images, which is inherently cumbersome and time-consuming, especially when managing images from numerous cameras or parking lots (Grbić & Koch, 2023; Nguyen & Sartipi, 2024). This process involves manually defining each parking slot area so that individual image patches can be extracted and subsequently classified as occupied or vacant. (Grbić & Koch, 2023).

Furthermore, this manual annotation procedure lacks robustness, as it must be entirely repeated if the camera angle shifts, if the camera is replaced, or if maintenance requires re-zooming the view (Grbić & Koch, 2023; Nguyen & Sartipi, 2024). This dependency on manual labeling severely impacts the

scalability and practicality of implementing vision-based parking guidance systems in real-world scenarios. Recognizing this limitation, the need for automatic parking space detection has been identified as a critical requirement for modern vision-based parking lot management systems (Grbić & Koch, 2023).

b) Environmental and Illumination Challenges

Real-time parking detection systems face severe challenges stemming from unpredictable environmental conditions and drastic lighting changes (Encío et al., 2023; Grbić & Koch, 2023). These systems must be robust enough to handle varying weather conditions such as snow, rain, and cloudy days, which significantly alter visual cues (Grbić & Koch, 2023; Yuldashev et al., 2023). Specifically, illumination changes between daytime and nighttime operation, including shadows cast by surrounding objects like buildings or trees, can introduce ambiguity that hinders detection accuracy.

For instance, certain traditional methods relying heavily on image processing techniques are notably inefficient across different weather conditions (Grbić & Koch, 2023). The presence of solar reflections from vehicles or glare on wet ground also complicates feature extraction, often leading to misclassifications.

Additionally, in low-light environments, the outlines of parking spaces or dark-colored cars may be difficult to discern or effectively threshold, contributing to increased false detection rates (Chen et al., 2023; Rahmat et al., 2024).

c) Computational Resource Constraints and Meeting Real-Time Speed

A critical problem in real-time parking detection is the substantial computational workload required, which often exceeds the capacity of

embedded or edge devices commonly used in autonomous vehicles or smart infrastructure (Amato et al., n.d.; da Luz et al., 2024; Popereshnyak & Chornobryvets, 2025). Parking Guidance Information (PGI) systems necessitate accurate and real-time information. However, achieving this high speed and accuracy often requires running complex models, like Deep Learning (DL) classifiers, which demand significant computing power and storage.

This computational demand presents a high barrier to achieving low latency, which is essential for providing instantaneous updates to drivers. Furthermore, methods that utilize complex, multi-stage detection pipelines, such as those relying on region proposal steps (like Faster R-CNN), may be too slow to satisfy the necessary real-time inference constraints for embedded platforms (Amato et al., n.d.; Grbić & Koch, 2023; Li et al., 2020).

Even synthesizing composite views, such as Around View Monitor (AVM) images, can introduce processing latency that negatively impacts real-time performance (Aajal et al., 2025).

d) False Detections from Transient or Illegal Vehicles

Distinguishing vehicles that are genuinely occupying a defined parking spot from those that are merely passing through or parked illegally remains a prominent challenge in real-time detection (Grbić & Koch, 2023). The presence of moving vehicles or illegally parked cars often leads to undesirable false positive detections, where an occupied status is incorrectly assigned to a vacant space. Object detection algorithms, especially those relying solely on visual presence (vehicle detection techniques), frequently misidentify transient vehicles or short stops as occupied parking slots (Nguyen & Sartipi, 2024).

This complexity requires the system not only to detect vehicles but also to implement sophisticated temporal analysis or tracking mechanisms to differentiate stationary parked cars from moving ones. Robust systems must inherently possess the capability to distinguish and ignore detections corresponding to illegal parking spots, bus stops, or passing vehicles to ensure the accuracy of occupancy reports (Grbić & Koch, 2023).

e) Occlusion and Generalization Across Diverse Viewpoints

Vision-based systems struggle when object occlusion prevents a clear view of the parking space or when the trained model is applied to a new environment with a different camera perspective. Occlusions can be caused by various elements such as trees, lampposts, adjacent vehicles, or other permanent structures in the parking area, making certain parking slots difficult or impossible to detect entirely (Grbić & Koch, 2023). Moreover, the challenge of generalizing a model trained under one set of viewing conditions (i.e., camera viewpoint, height, or angle) to function accurately across entirely different viewpoints is significant. If models exhibit weak generalization capabilities, they perform well only in the specific conditions used for training, necessitating expensive retraining and reconfiguration when transferred to new parking lots or different camera setups (Encío et al., 2023; Garta et al., 2025).

f) Handling Varied Geometries and Angular Orientation

A structural problem arises from the geometric complexity of modern parking layouts, requiring detection methods to accurately delineate slots that come in perpendicular, parallel, or slanted (diagonal) configurations (Encío et al., 2023; Wang et al., 2023).

Traditional bounding box representations are often insufficient, particularly for accurately representing diagonal slots or handling the varying orientations

caused by camera perspective. Without precise geometric modeling, the bounding boxes generated by detectors may inaccurately fit the visible area of the parking space, especially when partial occlusions or lens distortions are present (Encío et al., 2023; Wang et al., 2023; Xiao et al., n.d.).

This difficulty necessitates advanced strategies, such as using polygon-shaped representations or estimating orientation angles, to ensure that the prediction accurately captures the slot boundaries and location, rather than relying on simple axis-aligned rectangles (Wang et al., 2023).

g) **Advanced Classification for Suitability and Constraints**

Beyond merely classifying a parking spot as occupied or vacant, a significant area of advanced research involves identifying whether a free slot is compatible with the needs of the ego vehicle or suitable for specific user constraints, such as identifying spots designated for electric vehicles (EVs) or handicap accessibility. This classification task goes beyond basic occupancy detection and directly addresses the compatibility of a spot with the driver's current mission.

This requirement is crucial for developing context-aware parking guidance systems and autonomous valet parking scenarios. Researchers have addressed this by extending object detection algorithms, such as adapting YOLOv4, to classify parking spots into specific categories like regular, handicapped, or EV charging spots.

This detailed classification relies on sophisticated models capable of identifying visual cues within the parking area, and often utilizes specific bounding box formats, such as a four-point polygon model, which is better suited for various slot shapes (including slanted parking slots) than traditional rectangular boxes. By incorporating this advanced layer of classification, vision-based parking systems can provide more intelligent and context-aware guidance to drivers and autonomous systems (Xiao et al., n.d.).

h) Dataset Limitations

Existing publicly available datasets present several limitations that hinder the development of robust and generalizable real-time detection models. A primary concern is the **lack of crucial features and diversity**; for example, surveys indicate that available datasets frequently lack video sequences, and images collected under severe conditions such as nighttime operation or snow are vastly underrepresented (de Almeida et al., 2022).

While datasets like PKLot and CNRPark-EXT do include images under varying weather conditions (sunny, rainy, cloudy), none of them originally contained video sequences (de Almeida et al., 2022; Grbić & Koch, 2023). The scarcity of challenging visual data, such as images containing snow (only 1.7% in surveyed public data) or captured at night (only 12% in surveyed public data), prevents models from being adequately trained for general deployment.

Additionally, annotation discrepancies introduce problems; for instance, some regular parking slots might not be annotated within datasets like PKLot or CNRPark-EXT, potentially leading to **false positive detections** when testing automated detection algorithms (Grbić & Koch, 2023). Furthermore, some datasets rely on **non-rotated bounding boxes**, which fail to cover the parking slot area precisely or entirely when dealing with angled or diagonal slots (Amato et al., n.d.; Grbić & Koch, 2023).

Finally, a pervasive issue relates to the establishment of assessment protocols: many studies fail to account for the temporal aspect where the **same vehicle remains parked across numerous consecutive images**, leading to artificially inflated accuracy metrics (biased results) if the data split is performed randomly (de Almeida et al., 2022). Addressing these limitations requires establishing more complete and diverse datasets that include wider ranges of environmental variables, video data, and standardized labeling protocols (Wong et al.,

The synthesis of these problems and their evolving solutions forms the foundation for evaluating the current state-of-the-art and identifying critical research gaps, thereby providing the necessary context for the remainder of this chapter.

2.2 Reliance on Manual Labeling and Scalability Limitations (Automatic Parking Slot Detection Gap)

The necessity for manual labeling and annotation maintenance represents one of the most critical challenges to achieving scalable and practical real-time car park slot detection, creating a persistent gap in automatic parking space detection (APSD) capability (de Almeida et al., 2022; Grbić & Koch, 2023). This process requires extensive human effort to manually define the location of each parking slot within the input image, often by drawing bounding boxes or polygons, so that individual image patches can be accurately extracted for subsequent classification as either occupied or vacant (Grbić & Koch, 2023).

This manual setup is highly cumbersome and time-consuming, severely limiting the practicality and scalability of vision-based systems, especially when large numbers of cameras or extensive parking facilities are involved (Grbić & Koch, 2023). Furthermore, a significant problem is the non-robustness of these manual annotations; if factors such as the camera angle shift, the camera unit is replaced, or maintenance requires re-zooming the field of view, the entire labeling procedure must be fully repeated (Grbić & Koch, 2023).

The reliance on systems that demand *a priori* configuration information, such as estimating the camera homography matrix or manually inputting the corner coordinates and number of slots, also limits the generality of the approach and increases deployment complexity (Grbić & Koch, 2023).

Consequently, researchers have widely recognized that the ability to perform **automatic parking space detection** is a fundamental and critical

prerequisite for developing modern, scalable, and deployable vision-based parking lot management systems (de Almeida et al., 2022; Grbić & Koch, 2023). The core aim of solving this manual dependency is to allow the system to determine parking slot locations directly from input images, thereby accelerating implementation and ensuring robustness against small environmental or maintenance changes.

Researchers have proposed several comprehensive, automated approaches to eliminate the reliance on manual annotation and achieve automatic parking slot detection:

2.2.1 Vehicle Detection and Density-Based Clustering (APSD Approach):

A primary proposed solution involves determining slot locations by leveraging the temporal occurrence of parked vehicles (Grbić & Koch, 2023). This method relies on the assumption that vehicles consistently park within defined slots and typically follows a multi-step process:

- **Vehicle Detection and Data Collection:** An object detector (e.g., YOLOv5) is applied to a series of images captured over a period of time to detect and store the center coordinates of bounding boxes for all observed vehicles (Grbić & Koch, 2023).
- **Perspective Transformation:** The stored bounding box centers are mathematically **transformed into a bird's-eye view** using a homography matrix (often calculated by a dedicated Convolutional Neural Network or estimation techniques). This step minimizes the distortions caused by camera perspective and equalizes the perceived distance between adjacent parking slots, improving clustering efficiency (Grbić & Koch, 2023).

- **Density Clustering:** A density-based spatial clustering algorithm, such as DBSCAN, is applied to the projected data to identify **high-density clusters** that correspond to the consistent locations of marked parking slots (Grbić & Koch, 2023).
- **Cluster Filtering:** The detected cluster centers are rigorously **filtered** by analyzing the deviation or standard deviation of the bounding box centers around the cluster mean. This allows the algorithm to **discard transient or spurious detections** arising from passing vehicles, illegal parking spots, or detection noise, selecting only the locations corresponding to regular parking slots (Grbić & Koch, 2023).

2.2.2 Object Detection with Vehicle Tracking

This method uses a two-step approach where an object detector (like Faster R-CNN or YOLOv4) detects vehicles in real-time, followed by a tracking algorithm to differentiate between stationary, parked vehicles and moving vehicles (de Almeida et al., 2022; Grbić & Koch, 2023; Nguyen & Sartipi, 2024). Spots are marked based on where a vehicle remains stationary for a defined time or number of frames. This approach requires setting thresholds for intersection over union (IoU) and the minimum number of frames () to confirm a vehicle is parked (Nguyen & Sartipi, 2024).

2.2.3 Geometric Modeling with Advanced Representation

Instead of relying on simple axis-aligned bounding boxes (BBs), advanced models like the Holistic Parking Slot Network (HPS-Net) directly regress the coordinates of the **four vertices of a polygon** (quadrilateral) to represent the parking slot in a top-view domain (Wang et al., 2023). This polygon representation is robust enough to accommodate all parking shapes (perpendicular, parallel, diagonal) and accurately fit visible areas even when occlusion occurs, avoiding manual slot definition (Wang et al., 2023).

2.2.4 Heatmap Generation from Aggregated Detections

Utilizing deep learning networks (e.g., Cascade Mask R-CNN or segmentation models) to perform car segmentation or detection over numerous video frames (Nguyen & Sartipi, 2024). By aggregating the segmentation or detection results, a cumulative heatmap is generated that reliably delineates the spatial positions of frequently occupied parking slots based on a defined threshold (Nguyen & Sartipi, 2024).

2.2.5 Chess-Board/Grid Assumption

Assuming the parking area adheres to a simple rectangular or grid structure, the boundaries of each parking space can be defined automatically once corner coordinates, the total number of rows and columns, and a homography matrix estimation (to achieve bird's-eye view) are provided (de Almeida et al., 2022; Grbić & Koch, 2023; Nguyen & Sartipi, 2024).

2.2.6 Automatic Bounding Box Adjustment

Implementing mechanisms like the PakSke module, designed to automatically determine the optimal orientation angle, width scaling, and height scaling for bounding boxes detected in parking spots, ensuring alignment with the actual marked boundaries, thus removing the need for manual adjustment in varying camera perspectives (Nguyen & Sartipi, 2024).

2.3 Environmental and Illumination Challenges

Environmental and illumination challenges constitute one of the most persistent and complex difficulties in achieving reliable real-time car park slot detection, as vision-based systems must consistently perform across a wide spectrum of unpredictable outdoor and urban conditions. A fundamental obstacle is the requirement for vision-based systems to remain robust against significant illuminance changes during daytime and nighttime operation, a critical capability

for any deployable parking guidance information (PGI) system (Grbić & Koch, 2023).

Daytime operations are routinely hampered by unpredictable phenomena such as shadows cast by structures like buildings or trees, which introduce ambiguity that degrades detection accuracy (Amato et al., n.d.). Furthermore, intense sunlight can produce visual artifacts, including solar reflections from vehicles or glare on wet ground, which severely complicates the feature extraction process and often leads to misclassifications (Amato et al., n.d.).

Conversely, performance suffers drastically in low-light or nighttime environments, where traditional detection techniques struggle because the outlines of parking spaces or dark-colored cars may be difficult to discern or effectively threshold, resulting in increased false detection rates (Rahmat et al., 2024). This challenge is particularly acute in underground garages or weak lighting conditions where the lack of visible features hinders deep learning models (Chen et al., 2023).

Beyond illumination shifts, systems must contend with dynamic and unpredictable weather events, including snow, rain, and cloudy days, all of which significantly alter visual cues (Amato et al., n.d.; Grbić & Koch, 2023). Rainy weather introduces multiple detrimental factors: it generates glare on wet ground or reflections on a car's windscreen, covering parts of the images with saturated patterns, and water puddles and wet floors create confusing textural patterns that may lead to misclassification (Amato et al., n.d.; Grbić & Koch, 2023).

Traditional computer vision methods are notably prone to inefficiency when processing input across various weather conditions (Grbić & Koch, 2023; Li et al., 2020). The difficulty in handling these challenges is exacerbated by a pervasive data limitation: publicly available datasets often feature a critical scarcity of challenging visual data captured under extreme conditions, such as

images containing heavy snow or collected during nighttime operation (de Almeida et al., 2022; Wong et al., 2023).

This lack of diverse training data severely restricts the development of generalizable models capable of maintaining high accuracy and performance when deployed across varied real-world scenarios (Amato et al., n.d.; Grbić & Koch, 2023). Ultimately, successfully addressing these environmental and illumination variables is paramount to overcoming the limitation that traditional monitoring techniques are susceptible to change and ensuring that modern deep learning models can operate reliably in 24/7 urban settings (Amato et al., n.d.; Encío et al., 2023; Li et al., 2020). Solutions Discussed by Researchers to Solve Environmental and Illumination Challenges:

2.3.1 Robustness via Deep Learning and Data Augmentation

- The inherent capability of deep Convolutional Neural Networks (CNN) is leveraged to provide solutions that are robust to disturbances caused by shadows and the variation of light conditions (Amato et al., n.d.).
- The use of extensive **data augmentation** techniques during model training, such as randomly adjusting brightness/contrast, varying saturation, applying rotation, and performing scaling and horizontal flipping, enhances the models' robustness and generalization ability across different lighting and environmental conditions (Amato et al., n.d.; Popereshnyak & Chornobryvets, 2025; Shreeram et al., 2025; Wang et al., 2023; Wong et al., 2023).
- Models are trained or fine-tuned using large, challenging datasets (like CNRPark-EXT or PS2.0) that encompass diverse lighting and weather conditions to ensure resistance to environmental variations (Popereshnyak & Chornobryvets, 2025; Yuldashev et al., 2023).

- The customized **Deep CNN (DCNN) model** is intentionally designed to be resilient against **lighting variations and diverse circumstances**(Li et al., 2020).

2.3.2 Architectural Adaptations and Attention Mechanisms

- The proposed overall architecture VPS-Net combines a YOLOv3-based detector with a customized DCNN to robustly handle complicated visual environments, including light reflections, shadows, diverse ground materials, and dim street lighting (Li et al., 2020).
- Integrating the GELU (Gaussian error linear unit) activation function into advanced detection models (like improved YOLOv5-OBB) enhances generalization ability in complex scenarios such as darkness (Chen et al., 2023).
- The Coordinate Attention (CA) mechanism can be introduced into the network backbone to improve feature expression, helping to reduce the false detection rate and improve accuracy specifically under weak lighting conditions (Chen et al., 2023).
- The introduction of the RFB (Receptive Field Block) module serves to increase the receptive field, which consequently improves the detection accuracy of small targets (such as parking space corners) that might otherwise be missed due to occlusion or poor visibility (Chen et al., 2023).
- The Select-and-Prune (SP) Module is used in lightweight CNN models to optimize network complexity for real-time performance while maintaining high detection accuracy, particularly important when resource constraints limit the use of larger, more inherently robust models (Wong et al., 2023).

2.3.3 Advanced Feature and Contextual Processing

- Holistic (Multi-Branch Output) neural network approaches enhance robustness by utilizing the whole image's contextual information to infer occupancy, thereby handling illumination changes, occlusions, and perspective distortions better than methods relying solely on small patches (Encío et al., 2023).
- One study advocated replacing the commonly used ReLU6 activation function with LeakyReLU6 in deep learning models (like MobileNetV3) to prevent neurons from becoming inactive (dying) when encountering negative input values, leading to more effective feature extraction in low-light scenarios (Yuldashev et al., 2023).
- The semantic segmentation approach can be employed to explicitly delineate and segment static environmental factors (like parking slot lines and other static objects) to manage challenges related to occlusions and visual noise (Wong et al., 2023).
- Integrating Kernelized Correlation Filter (KCF) trackers allows the system to maintain real-time updates on marking points, augmenting robustness and reducing temporal noise caused by fluctuating visibility (Aajal et al., 2025).

2.3.4 Geometric and Temporal Filtering

- Motion detection algorithms are employed to differentiate stationary, parked cars from moving or transient vehicles, preventing misclassification due to passing traffic that might occur during periods of high visual noise (Francis et al., 2023).
- A method using background subtraction determines potential parking slots by monitoring temporal changes of parked cars, which is a technique robust against illumination issues where traditional painted markings become hard to distinguish (Encío et al., 2023).

- In the geometric context, implementing polygon or circular descriptors offers robustness to rotation and perspective distortions, increasing reliability compared to systems using less flexible rectangular descriptors (Aajal et al., 2025).

2.4 Computational Resource Constraints and Meeting Real-Time Speed

The challenge of Computational Resource Constraints and Meeting Real-Time Speed is a severe impediment to deploying vision-based car park slot detection, particularly when targeting low-power devices necessary for scalable smart city infrastructure.

Providing accurate and real-time information is essential for Parking Guidance Information (PGI) systems to function effectively (Grbić & Koch, 2023). However, achieving the required high speed and accuracy often necessitates employing complex models, such as Deep Learning (DL) classifiers or convolutional neural networks (CNNs), which demand significant computing power and storage (de Almeida et al., 2022; Grbić & Koch, 2023; Popereshnyak & Chornobryvets, 2025).

This computational requirement routinely surpasses the limited capacity of readily available embedded, distributed, or low computing-power devices like smart cameras or platforms such as the Raspberry Pi (Amato et al., n.d.; da Luz et al., 2024; Grbić & Koch, 2023; Popereshnyak & Chornobryvets, 2025; Wong et al., 2023; Yuldashev et al., 2023).

This mismatch between computational need and hardware capability directly results in high latency, severely undermining the systems' ability to provide instantaneous updates, a core requirement for guiding drivers or autonomous vehicles (Grbić & Koch, 2023; Nguyen & Sartipi, 2024; Popereshnyak & Chornobryvets, 2025). Several aspects of complex detection pipelines create specific performance bottlenecks:

First, systems relying on multi-stage detection methods, such as those incorporating region proposal steps (like Faster R-CNN), are typically too slow to meet the strict real-time inference constraints essential for deployment on embedded platforms (Grbić & Koch, 2023).

Second, complexity is compounded by the sheer volume of data processing required in many deployments. For instance, successfully implementing Deep Learning classifiers requires each monitored parking space to execute an independent occupancy detection task (Amato et al., n.d.). If a single smart camera is tasked with monitoring several parking spaces simultaneously, running complex architectures like the well-known AlexNet model on a low-computing device becomes a very difficult challenge due to the aggregate computational load (Amato et al., n.d.).

Furthermore, as the number of slots increases, running traditional classification methods demands numerous forward passes, leading to sluggish real-time performance (Nguyen & Sartipi, 2024). Third, latency is significantly introduced even before core detection begins in systems utilizing Around View Monitor (AVM) images (Aajal et al., 2025). The composite AVM image must be synthesized from multiple camera feeds by the vehicle's central processing unit (CPU) as a necessary preprocessing step, delaying the actual downstream analysis and affecting real-time performance (Aajal et al., 2025; Wang et al., 2023; Wong et al., 2023).

The drive to shift processing toward the sensor (Edge Computing) to reduce communication overhead and eliminate central server bottlenecks further emphasizes the need for lightweight models capable of running efficiently on resource-constrained hardware (Amato et al., n.d.; da Luz et al., 2024). Solutions Discussed to Solve Computational Resource Constraints and Meeting Real-Time Speed:

2.4.1 Decentralized and Edge Computing Deployment

Deploying the entire occupancy detection software directly on **smart cameras** (such as Raspberry Pi platforms) eliminates central server bottlenecks and reduces communication overhead, thereby improving system scalability (Amato et al., n.d.; Grbić & Koch, 2023; Yuldashev et al., 2023).

2.4.2 Adoption of Lightweight and Highly Efficient Architectures

- Designing **smaller deep CNN architectures** (e.g., mAlexNet, a reduced version of AlexNet) specifically tailored for execution on embedded, low computing-power devices, resulting in models that are significantly smaller (e.g., three orders of magnitude smaller than AlexNet) (Amato et al., n.d.; Grbić & Koch, 2023; Yuldashev et al., 2023).
- Utilizing **single-stage object detectors** (such as YOLOv3, YOLOv5s, YOLOv7-tiny, YOLOv8n, or SSD) which are inherently optimized for fast, real-time inference by eliminating multi-stage processes like region proposal generation (Chen et al., 2023; Garta et al., 2025; Grbić & Koch, 2023; Nguyen & Sartipi, 2024; Popereshnyak & Chornobryvets, 2025).
- Employing **MobileNet family models** (like MobileNetV3) known for low latency, small size, and high efficiency, making them suitable for mobile and embedded applications (Yuldashev et al., 2023).

2.4.3 Architectural Optimization and Customization

- Implementing techniques like the Select-and-Prune (SP) Module within networks (e.g., FCNs) to selectively identify essential receptive fields and prune redundant channels, optimizing the neural network for real-time speed on limited computational resources (Aajal et al., 2025; Wong et al., 2023).

- Substituting inefficient components, such as replacing standard convolutional layers with depth-wise separable convolutions or blueprint separable convolutions (BSCConv), to drastically reduce the model's number of parameters and computational complexity (Yuldashev et al., 2023).
- Introducing modern efficiency improvements, such as adding the Receptive Field Block (RFB) module to improve feature representation for small targets without heavily impacting speed, and replacing activation functions like SiLU with GELU for enhanced generalization and reduced parameter count in the backbone network (Chen et al., 2023).
- Incorporating Coordinate Attention (CA) mechanism in the backbone and feature extraction stages to enhance feature expression capability, reducing false detection rates, especially under weak lighting conditions, without excessive computational cost (Chen et al., 2023).
- Designing networks (like the improved YOLOv5-OBB) where redundant modules like the Focus and SSP structures are replaced or removed to reduce the overall computational workload and accelerate model inference (Chen et al., 2023).

2.4.4 Data and Pipeline Adjustments

- Leveraging transfer learning, which dictates that once a model is trained, the run-time phase of prediction requires significantly fewer computational resources than the initial training phase, making deployment on low-power devices feasible (Amato et al., n.d.; Grbić & Koch, 2023).
- Shifting the reliance away from computationally expensive image synthesis, such as utilizing single-camera bird's-eye view (BEV) images

instead of AVM images, enabling earlier processing in the pipeline and thus reducing latency and improving efficiency (Aajal et al., 2025).

- **Model Compression Techniques:** Exploring future work in model optimization techniques like quantization, conversion to TensorFlow Lite format, or model pruning to further reduce model complexity and improve runtime performance for embedded hardware deployment (Aajal et al., 2025; Chen et al., 2023; da Luz et al., 2024; Wong et al., 2023).

2.5 False Detections from Transient or Illegal Vehicles

The challenge of False Detections from Transient or Illegal Vehicles in real-time car park slot detection requires sophisticated mechanisms to accurately interpret visual data beyond mere vehicle presence, encompassing issues related to temporal persistence and spatial validity.

A core concern is the system's ability to differentiate vehicles genuinely occupying a designated parking slot from those that are merely passing through the camera's field of view, executing temporary stops, or parking illegally outside of marked areas (Grbić & Koch, 2023). When parking detection algorithms rely heavily on the visual output of vehicle detection techniques, they often generate undesirable false positive detections when transient vehicles are present, leading to the incorrect assignment of an occupied status to an otherwise vacant space (Grbić & Koch, 2023). For instance, loud or large vehicles passing through a monitored area may cause misclassification, requiring specialized mechanisms like a voting system to counteract these transient false positives (Grbić & Koch, 2023).

Furthermore, accurate detection systems must isolate designated parking spots from illegal or prohibited parking locations, such as sidewalks or entrances, which are not considered valid parking slots for occupancy reporting (Grbić & Koch, 2023). A basic object detection approach frequently

results in false positives due to illegally parked vehicles (Grbić & Koch, 2023). Systems utilizing vehicle detection in a sequence of images often observe low-density or outlying detections, often identified as "noise points," which typically stem from passing cars or short illegal stops (Grbić & Koch, 2023).

Resolving this ambiguity demands implementing spatial or temporal analysis, such as sophisticated tracking mechanisms, to confirm that a detected vehicle is stationary and positioned correctly within defined parking slot markings (Grbić & Koch, 2023). Another layer of complexity is introduced by the necessity of detecting small, non-vehicle obstacles (such targets are needed for full environmental awareness), such as ice cream cones, floor locks, or storage corners (Chen et al., 2023).

The small size of these objects often results in them being missed or causing errors when their bounding boxes overlap with parking spaces, leading to a decrease in overall detection accuracy (Chen et al., 2023). Consequently, a robust system must be capable of distinguishing regular parking slots from spatial anomalies or violations caused by transient activity or improper placement to ensure reliable occupancy reports (Grbić & Koch, 2023). Solutions Discussed to Solve False Detections from Transient or Illegal Vehicles:

2.5.1 Implement Density-Based Spatial Clustering (DBSCAN)

Apply clustering algorithms, such as DBSCAN, to the temporal collection of vehicle bounding box centers (often normalized to a bird's-eye view) to identify high-density clusters corresponding to consistently used, legitimate parking spots, thereby classifying low-density points as noise or outliers representing transient/illegal vehicles (Grbić & Koch, 2023).

2.5.2 Filter Cluster Centers by Deviation

Filter the calculated mean centers (m_i) of the clusters based on the deviation (s_i) of the bounding box centers around that mean, as vehicles parked legally within marked boundaries exhibit less deviation (low s_i) than illegally or haphazardly parked vehicles (high s_i) (Grbić & Koch, 2023).

2.5.3 Utilize Vehicle Tracking and Temporal Confirmation

Employ vehicle tracking across multiple consecutive frames to confirm if a detected object is stationary and parked, typically by requiring a consistently high Intersection over Union (IoU) value (e.g., $\theta = 0.75$) over a specified frame threshold ($\gamma = 4$) (Grbić & Koch, 2023; Nguyen & Sartipi, 2024).

2.5.4 Employ a Temporal Voting Mechanism

Use a voting mechanism or multi-frame fusion strategy to analyze temporal stability, allowing the system to ignore momentary false detections caused by large passing vehicles or transient events (Aajal et al., 2025).

2.5.5 Implement Location-Rule-Based NMS

Introduce a non-maximal suppression (NMS) optimization that incorporates a penalty factor (K) based on the modeled spatial correlation between objects (e.g., parking spaces and storage corners), improving detection confidence and accuracy, especially where spatial overlap exists between different categories (Chen et al., 2023).

2.5.6 Explicitly Exclude Non-Valid Parking Areas

Design the parking detection system to explicitly identify and disregard vehicle detections that correspond to known unauthorized locations, such as

sidewalks, bus stops, or entryways, ensuring that occupancy reporting is limited strictly to designated parking spaces (Grbić & Koch, 2023).

2.6 Occlusion and Generalization Across Diverse Viewpoints

A prevalent real-world problem is the physical occlusion of parking spaces or crucial parking boundary markings. Occlusions can stem from stationary infrastructure, such as lampposts, traffic signs, buildings, or large trees, as well as dynamic elements, most commonly adjacent parked vehicles or transient traffic passing through the view (Amato et al., n.d.; Encío et al., 2023; Grbić & Koch, 2023; Yuldashev et al., 2023).

These obstructions can hide partial or entire sections of a parking space, making it difficult or impossible for the detection system to confirm occupancy status accurately (Grbić & Koch, 2023). Specifically, partial occlusion patterns due to obstacles like trees, lampposts, and cars are frequently included as challenges in comprehensive datasets designed to test model robustness (Amato et al., n.d.). For models trained using image segmentation techniques (like Mask R-CNN), occlusions introduce noise that severely degrades accuracy (Aajal et al., 2025; de Almeida et al., 2022).

Furthermore, when cameras are mounted at lower points of view, such as sometimes seen in the CNRPark-EXT dataset, the likelihood of occlusion caused by adjacent vehicles significantly increases, intensifying the difficulty of the detection task compared to overhead views (Amato et al., n.d.). For any visual system aiming for round-the-clock operation, robustness against partial occlusions is a necessity (Amato et al., n.d.; Popereshnyak & Chornobryvets, 2025).

The Challenge of Generalization and Domain Shift

A central hurdle for deploying vision-based systems widely is the problem of generalization, which refers to a model's ability to maintain high

performance when applied to parking lots or scenarios that are significantly different from those used during the training phase (Amato et al., n.d.; Grbić & Koch, 2023). Historically, many traditional video-based techniques were meticulously tailored and fine-tuned to specific contexts, making their adaptation to a new parking lot extremely difficult (Amato et al., n.d.).

The core difficulty arises from the shift in domain caused by new visual factors, such as different lighting, weather, and especially changes in the camera's viewpoint, height, or angle (Aajal et al., 2025; Amato et al., n.d.; Grbić & Koch, 2023). If a classifier exhibits poor generalization, its high accuracy achieved in the training environment plummets when tested with images captured from a camera possessing a different point of view (Amato et al., n.d.). This reliance on highly specific conditions undermines the scalability and practical application of the system (de Almeida et al., 2022).

Impact of Perspective Distortion

Perspective distortion directly relates to generalization, as it is caused by the physical positioning of the camera relative to the parking lot. Cameras mounted at various angles inherently introduce geometric distortion, particularly noticeable in wide-angle or fisheye images (Aajal et al., 2025; Encío et al., 2023).

This visual distortion causes physical objects or parking slots that are farther away from the camera to appear smaller and closer together than those nearby, leading to discrepancies in detected feature densities and affecting the efficiency of processing algorithms (Grbić & Koch, 2023). This geometric variability, including oblique viewing angles, must be handled accurately, placing limitations on detection methods that rely on simple bounding box representations or fixed spatial assumptions (Aajal et al., 2025).

The distortion effects necessitate a fundamental preprocessing step to either mathematically correct the image perspective or utilize detection models that are inherently invariant to rotation and scaling to prevent misleading results (Aajal et al., 2025).

Solutions to these intertwined problems leverage advancements in deep learning architecture, sophisticated geometric modeling, and multi-sensor deployment strategies:

2.6.1 Holistic and Contextual Feature Utilization

- Employing Multi-Branch Output Neural Networks (MBONN) that process the common feature map extracted from the entire image using independent branches for each parking slot, enabling the system to adaptively analyze and predict occupancy robustly against mutual occlusions (Encío et al., 2023).
- Designing Deep Convolutional Neural Networks (DCNNs), such as mAlexNet or optimized ResNet architectures, specifically for robustness against visual disturbances like shadows and partial occlusions, lending the network better generalization properties (Amato et al., n.d.; Grbić & Koch, 2023).

2.6.2 Advanced Geometric Modeling and Correction

- Utilizing **Inverse Perspective Mapping (IPM)** to transform the original camera view into a normalized **bird's-eye view** (top view), thereby minimizing perspective distortion and equalizing the perceived distance and size of objects regardless of their location in the physical space (Aajal et al., 2025; Grbić & Koch, 2023; Wang et al., 2023).

- Automating the estimation of the **Homography Matrix (H)**, often using a dedicated CNN, to generalize the perspective transformation process and eliminate the time-consuming need for manual collection of top-view information for calibration (Grbić & Koch, 2023).
- Employing **Polygon-Shaped Representations (quadrilaterals)** instead of traditional axis-aligned bounding boxes to accurately model diverse slot shapes (perpendicular, parallel, slanted) and to precisely fit the visible area of **occluded slots** (Aajal et al., 2025; Wang et al., 2023).
- Incorporating **circular descriptors** to achieve inherent **rotation invariance** and robustness against changes in perspective distortion or angle, making the solution universally applicable across varied viewpoints (Aajal et al., 2025).

2.6.3 Enhanced Training and Data Strategies for Generalization

- Training models on **extensive, challenging datasets** (e.g., CNRPark-EXT, PS2.0) that intentionally include a high variability of images covering diverse viewpoints, occlusions, and environmental conditions to improve a model's ability to generalize to unseen data (Amato et al., n.d.; de Almeida et al., 2022).
- Evaluating models rigorously by training on one dataset or camera viewpoint and **testing on a completely different one** (inter-camera/inter-dataset evaluation) to genuinely test the generalization property (Amato et al., n.d.; de Almeida et al., 2022; Grbić & Koch, 2023).
- Using **Augmentation Techniques** during training, such as random rotations, scaling, and perspective shifts, to artificially increase the complexity and diversity of the training data and enhance the model's resistance to input variations in real-world deployment (Popereshnyak & Chornobryvets, 2025; Wong et al., 2023).

2.6.4 Temporal and Multi-View Solutions for Robustness

- Deploying redundancy through overlapping coverage from multiple cameras, and integrating a server-side mechanism to select the highest weighted confidence value from the camera offering the clearest, least-occluded view of a parking space, thereby effectively managing occlusion problems (Amato et al., n.d.).
- Implementing vehicle tracking using methods like Kernelized Correlation Filters (KCF) or temporal analysis to maintain positional accuracy and continuously refine the detection and localization of marking points across multiple video frames, enhancing robustness against transient occlusions or noise (Aajal et al., 2025).
- Modeling the parking space as a volume in 3D space (using hierarchical Bayesian frameworks) to explicitly account for occlusions when calculating the probability of a vehicle being present (Amato et al., n.d.).

2.7 Handling Varied Geometries and Angular Orientation

The necessity of accurately handling varied geometries and angular orientation stands as a paramount technical hurdle in achieving robust and generalized real-time car park slot detection. The problem originates from the fundamental geometric diversity found in modern parking environments, which routinely feature slots configured in perpendicular, parallel, and slanted (diagonal or fishbone) arrangements (Li et al., 2020; Wang et al., 2023; Xiao et al., n.d.).

Traditional detection approaches, particularly those reliant on simple, axis-aligned rectangular bounding boxes (BBs), prove highly inaccurate because they fail to cover the area of diagonal slots precisely or entirely (Amato et al., n.d.; Wang et al., 2023). This deficiency is compounded in real-world deployments by camera systems viewing the parking lot from various oblique

angles, which introduces perspective distortion (Grbić & Koch, 2023; Wang et al., 2023). This distortion visually skews the geometry, causing slots farther from the camera to appear closer together, thereby hindering the efficiency of classification or clustering algorithms designed for automatic localization (Grbić & Koch, 2023).

For advanced applications like autonomous driving, accurate spatial representation must go beyond simple occupancy status. Systems must precisely delineate the shape using all four corners and, crucially, accurately determine the orientation angle of the slot and identify the specific entrance line (Wang et al., 2023; Xiao et al., n.d.). Without an explicit mechanism to predict this orientation or align the prediction box to the slot's true bank angle, the detection results remain unreliable for downstream functions like path planning (Chen et al., 2023; Nguyen & Sartipi, 2024; Wang et al., 2023).

Furthermore, perspective distortion mandates that initial image acquisition—especially from fisheye or around-view monitor (AVM) cameras—undergo a corrective processing step, such as Inverse Perspective Mapping (IPM), to translate distorted coordinates into a uniform bird's-eye view where true spatial relationships are normalized (Chen et al., 2023; Nguyen & Sartipi, 2024; Wang et al., 2023; Xiao et al., n.d.).

Researchers have developed several strategies to address the representation and distortion challenges presented by diverse parking geometries:

2.7.1 Polygon-Shaped Representations

Employ **four-point polygon representations** (quadrilaterals) instead of traditional rectangular bounding boxes. This representation effectively models all types of slots, including **perpendicular, parallel, and slanted (diagonal) configurations**, and accurately fits the visible area of slots that are partially occluded or geometrically complex (Wang et al., 2023; Xiao et al., n.d.).

2.7.2 Polygon-Corner GloU Loss Function

Introduce specialized loss functions, such as the **polygon-corner Generalized Intersection over Union (GloU) loss**, specifically optimized for training models that predict the four polygon vertices. This loss enhances **slot orientation estimation** and ensures that the critical **entrance line** is correctly identified relative to the ending line by preserving the order of the predicted corners (Wang et al., 2023).

2.7.3 Automatic Bounding Box Skewness Adjustment

Implement post-processing modules, such as the **PakSke** module, to automatically adjust the rotation, width, and height scaling of detected bounding boxes. This ensures the output is precisely aligned with the actual angle of the parking spot, correcting for cases where slots are not perpendicular to the image frame (Nguyen & Sartipi, 2024).

2.7.4 Inverse Perspective Mapping (IPM) for Normalization

Utilize IPM or homography matrix estimation (sometimes achieved using a dedicated CNN) to transform coordinates from the camera's perspective into a **bird's-eye view (topview)**. This essential preprocessing step normalizes geometric relationships and eliminates **perspective distortion**, enabling more reliable downstream analysis like clustering (Chen et al., 2023; Grbić & Koch, 2023; Nguyen & Sartipi, 2024; Wang et al., 2023; Xiao et al., n.d.).

2.7.5 Orientation-Aware Deep Learning Models

Design networks, such as Directional Marking-Point Regression for Parking Slot Detection (DMPR-PS), that integrate directional marking points and explicitly regress the orientation angle of the parking slot (e.g., predicting and) to overcome rotation sensitivity inherent in traditional bounding box methods (Chen et al., 2023; Wang et al., 2023; Wong et al., 2023).

2.7.6 Circular Descriptors and Rotation Invariance

Adopt methods that employ circular descriptors or templates, providing an inherently rotation-invariant feature that is robust to varying orientations and perspective changes across different parking slot shapes (e.g., trapezoid, oblique, brick shapes) (Wang et al., 2023; Wong et al., 2023).

2.7.7 Angle Classification via Circular Smooth Labels (CSL)

To mitigate the angular boundary problem common in regression (where, for example, -90 degrees and 89 degrees are physically close but mathematically distant), researchers convert continuous angle prediction into a discrete classification problem using CSL techniques (Chen et al., 2023).

2.8 Advanced Classification for Suitability and Constraints

The challenge of **Advanced Classification for Suitability and Constraints** moves beyond merely answering the binary question of whether a parking slot is occupied or vacant, aiming instead to determine if a free space is actually compatible with the **needs or "mission of the ego vehicle"** (Xiao et al., n.d.). This sophisticated capability is increasingly critical for developing truly intelligent autonomous valet parking (AVP) and advanced driver-assistance systems (ADAS) (Xiao et al., n.d.).

A major problem addressed by researchers is the necessity of identifying slots designated for specialized use based on specific constraints (Xiao et al., n.d.). For instance, a vehicle may require a spot marked for **handicap accessibility** or one equipped as an **electric vehicle (EV) charging spot** (Xiao et al., n.d.). Fulfilling the vehicle's mission necessitates recognizing the visual cues associated with these specific categories within the parking area (Xiao et al., n.d.). This classification layer demands that the detection system is robust enough to process contextual visual information accurately to

classify slots into categories like **regular, handicapped, or EV charging spots**.

Furthermore, achieving this precise classification introduces significant technical complexities related to accurately representing the slot's geometry (Xiao et al., n.d.). Standard detection methods often rely on simple rectangular bounding boxes, which are frequently inadequate for depicting the actual shape and orientation of diverse parking configurations, such as perpendicular, parallel, and particularly **slanted (fishbone) parking slots (Xiao et al., n.d.)**. Accurate suitability determination requires methods capable of generating a precise delineation of the parking space. Addressing this challenge involves adopting highly flexible geometric models, such as the **four-point polygon bounding box model (quadrilateral)**, which can accurately fit the boundaries of all potential slot shapes.

This polygon representation is crucial for ensuring that the predicted boundaries align perfectly with the defined parking spot, allowing the system to confidently verify if a space meets specific geometric requirements for vehicle or user type. The detailed study of parking spot detection and classification utilizing **surround view camera systems (SVS)** for auto valet parking scenarios is considered a **pioneering effort** in addressing these comprehensive constraints (Xiao et al., n.d.).

Solutions Discussed to Solve Advanced Classification for Suitability and Constraints:

2.8.1 Adaptation of Object Detection Networks

The core solution involves adapting established neural network architectures, such as the **YOLOv4 algorithm**, to extend their output capability from basic localization and binary occupancy prediction to include multi-category classification for suitability (Xiao et al., n.d.).

2.8.2 Implementation of Polygon Bounding Box Model

Researchers employ a **novel polygon bounding box model** defined by four vertex coordinates (a quadrilateral) to represent the parking slot. This representation is specifically designed to accurately fit all possible slot shapes (perpendicular, parallel, and slanted/fishbone), addressing the limitations of traditional rectangular boxes when precise geometry is required for classification (Xiao et al., n.d.).

2.8.3 Explicit Classification Training

The system is specifically trained to classify vacant slots into relevant categories such as **regular, handicapped, and electric vehicle (EV) charging spots** (Xiao et al., n.d.).

2.8.4 Use of Customized Loss Functions

To train models for simultaneous localization and classification, specialized loss functions are used. The total loss function incorporates binary cross-entropy loss (L_{bce}) for the classification task alongside polygon regression loss ($L_{polygon_IoU}$) and objectless loss (L_{obj}) (Xiao et al., n.d.).

2.8.5 Input Processing via Surround View Systems

The visual input relies on the **surround view camera system (SVS)**, utilizing images from multiple fisheye cameras. This input is pre-processed using **Inverse Perspective Mapping (IPM)** to create a holistic, corrected bird's-eye view image, which provides the necessary context for complex classification tasks (Xiao et al., n.d.).

2.8.6 Focus on Auto Valet Scenarios

The research focuses specifically on the context of **auto valet parking scenarios**, validating that the advanced classification approach is effective for guiding autonomous vehicles based on compatibility needs (Xiao et al., n.d.).

2.9 Dataset Limitations

The effective development and rigorous evaluation of real-time car park slot detection systems are significantly hampered by several pervasive **Dataset Limitations**, spanning issues of dataset completeness, fidelity of annotations, and methodological flaws in testing procedures. A primary concern is the extreme **scarcity of data diversity and crucial features** within publicly available benchmarks.

Surveys reveal that existing datasets often severely lack **video sequences** (relying mostly on static images) and dramatically **underrepresent images collected under severe visual conditions**, such as heavy snow or full nighttime operation (de Almeida et al., 2022; Wong et al., 2023). Specifically, only the PLds dataset contains images collected in snow (representing a mere 1.7% of surveyed public data) and nighttime conditions (only 12% of surveyed public data) (de Almeida et al., 2022).

This profound lack of challenging, diverse visual data prevents researchers from adequately training models to be robust and reliable when deployed in general, 24/7 real-world urban environments (de Almeida et al., 2022). Beyond environmental data, most current datasets prioritize the foundational data points—positions, status, and climate conditions—but **lack labels for non-parking-related objects**, such as pedestrians, general obstacles, or granular segmentation masks for vehicles, which would be necessary for achieving comprehensive environmental awareness (de Almeida et al., 2022; Wong et al., 2023). Furthermore, some existing datasets

cover only a **limited physical range** (e.g., PS2.0 only covering 10m x 10m), which restricts the complexity of visual challenges presented to the models (Wang et al., 2023).

The **quality and format of annotations** present another key area of limitation. Accuracy issues arise because several popular datasets employ sub-optimal bounding box conventions; for example, the CNRPark-EXT dataset uses **non-rotated square masks** which frequently fail to align precisely with the parking slot area, potentially including irrelevant background space or excluding necessary information (Amato et al., n.d.; de Almeida et al., 2022; Grbić & Koch, 2023). Conversely, while the PKLot dataset uses rotated rectangular masks for more precise coverage, this level of detailed annotation is still lacking across large portions of public data (Amato et al., n.d.).

A significant challenge also lies in **annotation completeness**, as some ground truth files are known to omit certain **regular parking slots**, which can inadvertently introduce false positive errors during automated detection evaluations (Grbić & Koch, 2023). Furthermore, datasets are overwhelmingly designed for binary occupancy classification (occupied/vacant), omitting the **advanced classification** tags necessary for systems that must identify slot suitability based on constraints like handicap accessibility or EV charging capabilities (de Almeida et al., 2022).

Finally, widespread issues exist concerning **methodological rigor and standardization** in testing. Many studies rely on test protocols that fail to account for the **temporal correlation** inherent in parking lot data, where the exact same vehicle may remain parked across numerous consecutive image captures (de Almeida et al., 2022). By employing simple random data splits, models frequently encounter images of the same parked car in both training and testing sets, leading to **artificially inflated accuracy metrics** and assessments that reflect trivial learning rather than true generalization ability

(de Almeida et al., 2022). This deficiency underscores the critical need for standardizing both the data annotation quality and the evaluation protocols used across the research community (de Almeida et al., 2022).

Solutions Discussed by Researchers to Solve Dataset Limitations:

2.9.1 Develop New, Comprehensive Datasets

Researchers must focus efforts on establishing new publicly available datasets that are **more complete, challenging, and diverse** (de Almeida et al., 2022; Wong et al., 2023).

2.9.2 Include Video Sequences and Temporal Data

Future datasets should systematically collect and include **video sequences** rather than only static images to support models that rely on spatio-temporal analysis and object tracking (de Almeida et al., 2022; Wong et al., 2023).

2.9.3 Gather Data Under Extreme Conditions

New datasets must contain substantial volumes of images captured under **severe visual conditions**, such as during full **nighttime operation** or in **heavy snow**, to ensure model robustness and generalization (de Almeida et al., 2022; Wong et al., 2023).

2.9.4 Improve Annotation Quality and Detail

Future research efforts should prioritize defining **standardized labeling protocols** and improving annotation quality, including the adoption of precise geometric representations, such as polygon-shaped boxes or rotated bounding boxes (de Almeida et al., 2022; Wang et al., 2023).

2.9.5 Expand Labeling Scope

Datasets should incorporate **labels for non-slot objects** in the scene, such as segmentation masks for vehicles, pedestrians, and environmental obstacles, to enhance contextual awareness and detection fidelity (de Almeida et al., 2022; Wong et al., 2023).

2.9.6 Establish Advanced Annotation

For future autonomous systems, methods should explore **3D annotation techniques** instead of relying solely on 2D image or top-view domains, potentially using data derived from Lidar or high-definition (HD) maps (Wang et al., 2023).

2.9.7 Adopt Date-Based Split Protocols

Implement and enforce **date-based data split protocols** during evaluation to ensure that all images related to a specific parked vehicle remain confined exclusively to either the training or testing set, thus preventing artificial inflation of accuracy (biased results)(de Almeida et al., 2022).

2.9.8 Promote Generalization Testing

Encourage researchers to explicitly test model generalization capacity by adopting protocols that mandate **training on one dataset and testing on a completely different one** (e.g., PKLot for training, CNRPark-EXT for testing) (de Almeida et al., 2022).

2.9.9 Standardize Metrics and Protocols

Develop and use **standard test protocols** that specify data splitting and mandate the use of objective evaluation metrics (like F1-score or AUC) to enable

fairer, standardized comparisons between different methods (de Almeida et al., 2022).

2.9.10 Increase Transparency and Reproducibility

Authors should be strongly encouraged to make their **trained models and data publicly available** to facilitate replication and validation of reported results across the scientific community (Amato et al., n.d.; de Almeida et al., 2022).

2.10 Summary

The literature review reveals that the advancement of real-time car park slot detection systems, while rapidly progressing, is fundamentally challenged by a set of interconnected issues spanning data management, model complexity, environmental robustness, and geometric fidelity. The path toward deploying scalable and reliable vision-based solutions requires overcoming the persistent dependency on human effort and mitigating the inherent unpredictability of real-world environments.

The most critical barrier to scalability remains the immense effort associated with manual labeling of parking slot locations, a cumbersome and time-consuming process that must be entirely repeated if the camera perspective or configuration changes (Grbić & Koch, 2023). Researchers universally agree that the development of Automatic Parking Slot Detection (APSD) systems is mandatory. Effective solutions leverage time-series analysis of vehicle detections, often using density-based spatial clustering (DBSCAN) on data normalized into a bird's-eye view via Perspective Transformation, thereby automatically generating slot locations and dramatically reducing human labor (Grbić & Koch, 2023; Wang et al., 2023).

This drive for automation is paralleled by the need to ensure computational efficiency on resource-constrained embedded systems, such as those relying on

low computing-power devices like the Raspberry Pi (Amato et al., n.d.; Grbić & Koch, 2023). Achieving necessary real-time speeds mandates the utilization of lightweight, single-stage object detectors (e.g., YOLOv5s, YOLOv7-tiny, mAlexNet) which bypass the computational overhead of multi-stage pipelines (like Faster R-CNN) (Amato et al., n.d.; Grbić & Koch, 2023; Li et al., 2020; Popereshnyak & Chornobryvets, 2025).

Further efficiencies are gained through highly optimized architectures that incorporate techniques such as network pruning, specialized feature aggregation modules (SPPF, C2f, RFB), and efficient convolution mechanisms (BSConv) to reduce parameter count and increase inference speed (Aajal et al., 2025; Chen et al., 2023; Yuldashev et al., 2023).

Achieving consistent robustness against environmental and illumination challenges is paramount for continuous, 24/7 operation. Unpredictable factors like heavy shadows, solar glare, rain, snow, and low-light conditions hinder detection accuracy (Grbić & Koch, 2023). Deep Convolutional Neural Networks (DCNNs) offer built-in resilience, augmented by training techniques utilizing extensive data augmentation (e.g., adjusting HSV color spaces and brightness) (Amato et al., n.d.; Popereshnyak & Chornobryvets, 2025; Wang et al., 2023). For difficult low-light scenarios, models incorporate specialized attention mechanisms, such as the Coordinate Attention (CA) module, or optimized activation functions like LeakyReLU6 to ensure accurate feature extraction when visual cues faint (Chen et al., 2023; Yuldashev et al., 2023).

Model reliability is further secured by mitigating transient visual noise and generalization failures. False detections arising from transient or illegal vehicles are filtered using temporal stability verification—comparing bounding box location across consecutive frames using Intersection over Union (IoU) tracking or utilizing density clustering (DBSCAN) to classify non-stationary outliers as noise (Grbić & Koch, 2023; Nguyen & Sartipi, 2024). Additionally, poor generalization across diverse camera viewpoints due to physical occlusion or perspective

distortion is overcome by leveraging DCNN architectures specifically designed for robustness to partial occlusions, often combined with deploying redundant multi-camera systems that select the clearest view (Amato et al., n.d.; Grbić & Koch, 2023).

The complexity of modern parking layouts introduces a demand for high geometric precision to handle varied geometries and angular orientation (slanted, perpendicular, parallel configurations) (Wang et al., 2023). This necessitates a shift from inaccurate rectangular bounding boxes to flexible polygon-shaped representations (quadrilaterals defined by four vertices) to accurately fit irregular or occluded slot areas (Wang et al., 2023). These advanced representations require sophisticated training methods, including the use of specialized loss functions like the polygon-corner GloU to correctly order corner vertices and accurately predict the entrance line, and implementing modules like PakSke for automated orientation adjustment (Nguyen & Sartipi, 2024; Wang et al., 2023).

This geometric accuracy is foundational for pursuing Advanced Classification for Suitability and Constraints, allowing systems to classify vacant spots not merely as occupied or vacant, but according to purpose (e.g., handicap or EV charging status), a critical requirement for modern autonomous systems (Xiao et al., n.d.).

Finally, future progress depends significantly on addressing systemic dataset limitations (de Almeida et al., 2022). A widespread issue is the reliance on flawed evaluation protocols using random data splits that inadvertently mix temporally correlated images of the same parked vehicle into both training and testing sets, leading to artificially inflated accuracy metrics (de Almeida et al., 2022). Researchers advocate strongly for adopting date-based data split protocols and requiring more publicly available video sequences and images captured under challenging conditions (nighttime, snow) to ensure truly rigorous and meaningful model assessment (de Almeida et al., 2022).

In summary, the literature review underscores a clear movement toward holistic, end-to-end learning frameworks that integrate automated spatial detection, refined geometric modeling, and sophisticated contextual classification within highly efficient, lightweight deep learning architectures, ultimately striving for reliable, real-time performance on embedded automotive or edge computing platforms (Aajal et al., 2025).

Chapter 3: Requirements Analysis

3.1 Overview

This chapter presents the requirements analysis for the proposed vision-based parking slot monitoring system. The objective of this chapter is to identify user needs, current parking issues and system expectations in order to derive a set of requirements that guide the design and development of the proposed solution.

Based on the problem identified in Chapter 1 and the insights gained from the literature review in Chapter 2, parking management within campus environments presents both technical and user-related challenges. While existing studies primarily focus on detection accuracy and system performance, user satisfaction and real-world usability remain important considerations for an effective parking solution. Therefore, this chapter combines user feedback with system-oriented observations to ensure that the proposed system addresses both practical and technical requirements.

To achieve this, fact-finding techniques such as surveys are employed to gather data from parking users within the Multimedia University (MMU) campus. These techniques are used to understand current parking issues, user satisfaction levels and expectations for an improved parking system. The findings from this process form the basis for defining functional, non-functional and user requirements.

The outcomes of this chapter serve as a foundation for the system design presented in Chapter 4, ensuring that the proposed parking slot monitoring system is aligned with real user needs and realistic operational conditions.

3.2 Fact-Finding Techniques

Fact-finding techniques are used to collect information required for identifying system and user requirements. In this project, a questionnaire is

adopted to gather structured and quantitative feedback from parking users within the MMU campus.

Given that parking availability directly affects students and staff on a daily basis, collecting feedback from actual users is essential to understanding the limitations of the current parking situation and identifying features required for an optimal parking system. The selected techniques allow the project to capture both measurable trends and individual perspectives related to parking experiences.

3.2.1 Justification

A questionnaire-based approach was selected as the primary fact-finding technique for this project to systematically gather user perspectives on the current parking situation within the Multimedia University (MMU) campus. Parking availability and management directly affect a large population of students and staff, making it essential to collect feedback from a broad and diverse group of users.

The use of a questionnaire allows data to be collected efficiently from a relatively large sample size within a limited time frame, which is suitable for the constraints of a Final Year Project. By targeting approximately 60 respondents, with an ideal target of up to 100 participants, the survey enables the identification of common issues, user satisfaction levels and expectations for an improved parking system.

In addition, a structured questionnaire provides consistency across responses, allowing quantitative analysis of user needs and perceptions. This is particularly important for understanding recurring problems such as parking availability, time spent searching for parking spaces and user acceptance of automated or vision-based parking monitoring solutions.

The collected data can be easily analyzed and visualized using built-in tools provided by online survey platforms. Therefore, a questionnaire-based

approach is considered the most appropriate and practical method for gathering requirements data for the proposed parking slot monitoring system.

3.2.2 Questionnaire Design

The questionnaire was designed to collect structured and reliable feedback regarding the current parking experience at Multimedia University (MMU) and to assess the potential acceptance of a computer vision–based smart parking solution. The design focused on obtaining quantitative data that could be easily analyzed and visualized, while ensuring that the survey remained concise and user-friendly to encourage participation.

The survey was created and distributed using Google Forms and consisted primarily of closed-ended questions. This approach was chosen to ensure consistency in responses and to allow for clear comparison and statistical analysis across respondents. Only one optional open-ended question was included at the end of the questionnaire to capture additional suggestions that may not have been covered by the predefined options.

The questionnaire was structured into several logical sections, each serving a specific purpose:

a. Demographic Information

This section collected basic background information such as the respondent's role at MMU, faculty, and vehicle ownership status. These questions aimed to provide context for the responses and to ensure that feedback was obtained from individuals who actively experience parking conditions on campus.

b. Parking Behavior and Experience

Questions in this section focused on parking frequency, peak difficulty times, commonly used parking areas, and areas where respondents most frequently struggle to find parking. The aim was to identify patterns in parking usage and to determine specific locations and time periods where parking issues are most prominent.

c. Parking Issues and Improvement Needs

Multiple-selection questions were used to identify common parking-related problems, such as long search times or lack of availability information, as well as desired improvements. These questions were designed to capture the most critical pain points and user expectations without requiring lengthy written responses.

d. System Perception and Acceptance

Likert-scale questions were included to measure satisfaction with the current parking system, the importance of knowing real-time parking availability, and the perceived usefulness of a vision-based parking solution. These questions aimed to assess user readiness and acceptance of the proposed system.

e. Parking Application Experience

Conditional questions were included to gather information about prior usage of parking applications and the features that respondents found most useful. This helped inform us of the functional requirements of the proposed system.

In general, the questionnaire was designed to balance depth and simplicity, ensuring that each question directly contributed to understanding user needs, system feasibility and practical implementation considerations for the proposed smart parking solution.

3.2.2.1 Analysis on results

The analysis of the questionnaire responses was conducted using a Jupyter Notebook environment to ensure systematic, reproducible and transparent data processing. Python was selected as the primary analysis tool due to its extensive support for data analysis and visualization. The libraries used in this analysis include **Pandas** for data cleaning and manipulation, **Matplotlib** and **Seaborn** for data visualization.

All survey responses collected through Google Forms were first exported into a spreadsheet format and subsequently imported into the Jupyter Notebook for analysis. Initial data preparation involved removing incomplete entries, standardizing categorical responses and mapping numerical Likert-scale values to descriptive labels to improve interpretability in the visual outputs.

To enhance the clarity of the analysis and focus on meaningful trends, a response frequency threshold was applied. Response options that were selected by fewer than four respondents were excluded from graphical representation. This threshold was introduced to reduce visual noise, avoid over-emphasizing outliers and ensure that the analysis highlights dominant patterns that are representative of the respondent population. Such an approach is commonly used in descriptive survey analysis to improve readability and analytical relevance.

The processed data was then analyzed question by question, with each result presented using appropriately selected chart types such as bar charts and horizontal bar charts. Visualizations were designed with consistent styling, clear labelling and value annotations to support straightforward interpretation. The findings presented in the following subsections are based solely on the analyzed survey data and form the foundation for subsequent observations and requirement identification.

a. Question 1: Role at MMU

Figure 3.1 illustrates the distribution of respondents based on their role at Multimedia University (MMU). The majority of respondents were undergraduate students, accounting for 63 responses. This was followed by visitors with 9 responses, postgraduate students with 8 responses and staff members with 5 responses.

The results indicate that most of the survey participants are students who regularly access the campus and are therefore directly affected by parking availability and congestion issues.

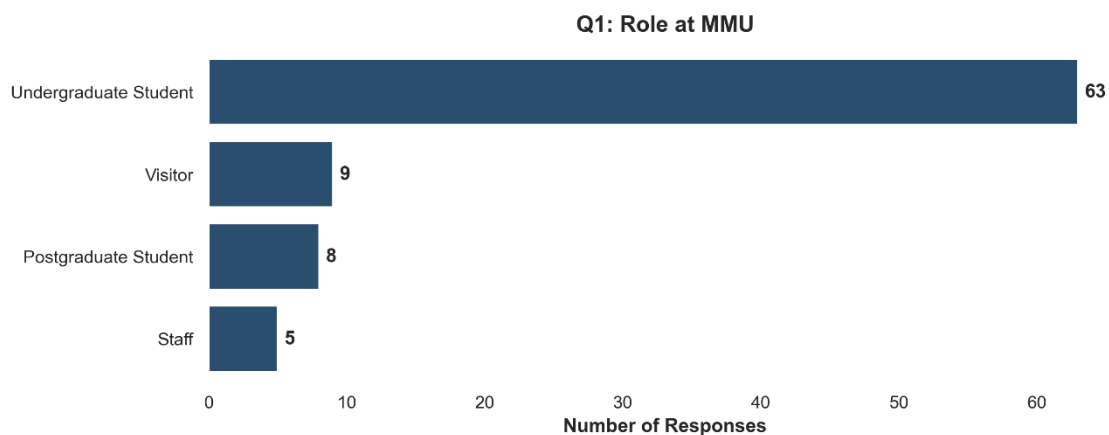


Figure 3.1: Respondents Role

b. Question 2: Faculty

Figure 3.2 presents the distribution of respondents by faculty. The Faculty of Computing and Informatics (FCI) recorded the highest number of responses with 44 participants. This was followed by the Faculty of Creative Multimedia (FCM) with 11 responses, the Faculty of Artificial Intelligence and Engineering (FAIE) and the Faculty of Management (FOM) with 10 responses each. A smaller

number of respondents selected “Does not Apply” (5 responses) and the Faculty of Applied Communications (FAC) (4 responses).

This distribution shows that responses were obtained from multiple faculties across MMU, providing a diverse representation of the campus community.

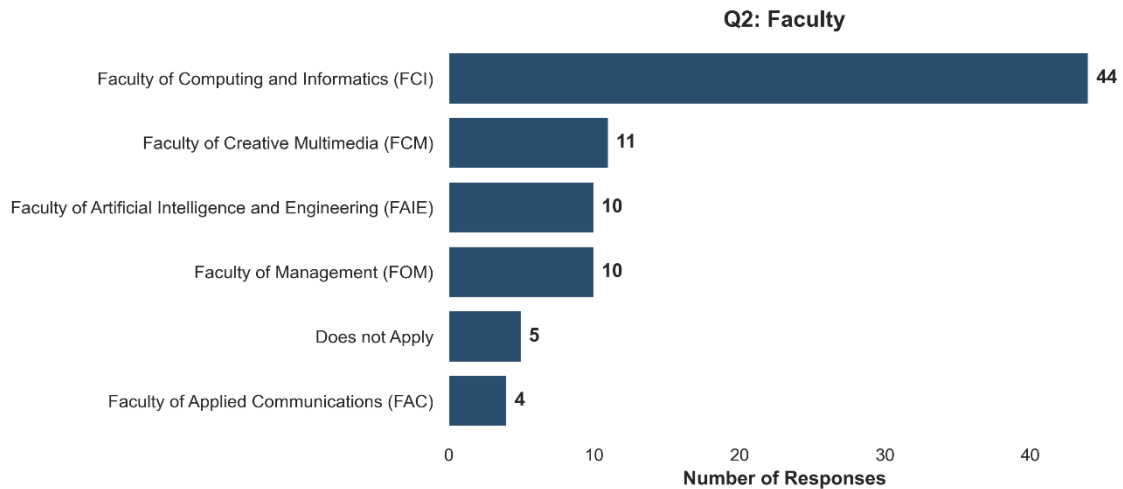


Figure 3.2: Respondents Faculties

c. Question 3: Vehicle Ownership

Figure 3.3 illustrates the vehicle ownership status of the respondents. The results show that a majority of respondents, representing **65%**, indicated that they own a vehicle, while **35%** reported that they do not own a vehicle. For more context, the 35% who do not own a vehicle, usually carpool with their friends to campus.

This distribution highlights that most participants have direct and regular exposure to parking facilities within the MMU campus, making their feedback particularly relevant for evaluating parking-related challenges and potential system improvements.

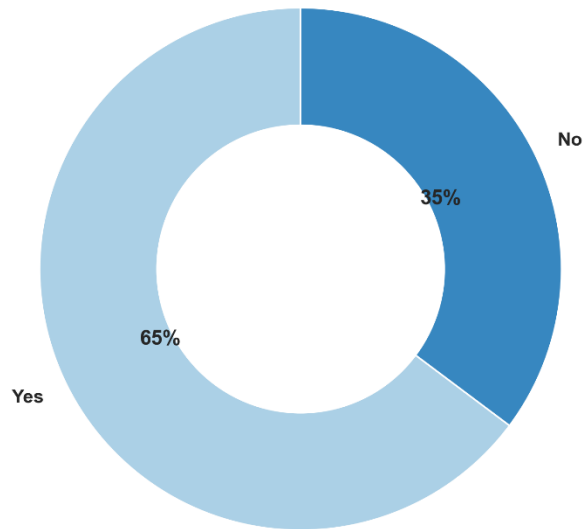
Q3: Vehicle Ownership*Figure 3.3: Vehicle Ownership***d. Question 4: Parking Frequency**

Figure 3.4 presents the frequency with which respondents park their vehicles on campus. The highest number of respondents reported parking **rarely** (31 responses), followed by those who park **3–4 times per week** (27 responses). A substantial number of respondents indicated that they park **daily** (22 responses), while a smaller group reported parking **1–2 times per week** (8 responses).

The 31 respondents who reported rarely, have expressed their dissatisfaction with the current parking issues multiple times, which eventually led them to not using MMU's parking spaces.

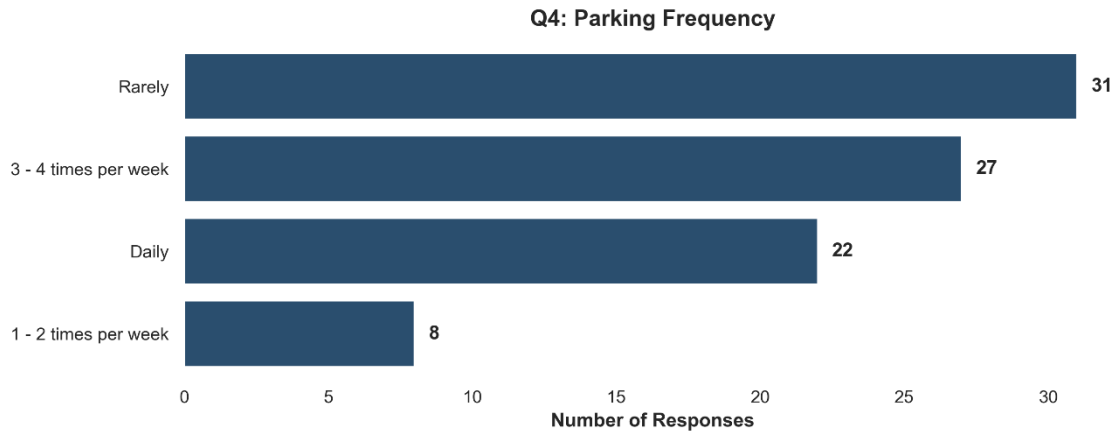


Figure 3.4: Parking Frequency

e. Question 5: Time of Highest Parking Difficulty

Figure 3.5 illustrates the time period during which respondents experience the highest difficulty in finding parking on the MMU campus. The highest number of respondents (45 responses) identified **midday (10:00 a.m. – 2:00 p.m.)** as the most challenging time for parking. This was followed by the **morning period (8:00 a.m. – 10:00 a.m.)** with 17 responses. A notable group of respondents (15 responses) indicated that they rarely face parking difficulties, while the **afternoon period (2:00 p.m. – 5:00 p.m.)** recorded the lowest number of responses (10).

The high concentration of parking difficulty during midday suggests a peak accumulation of vehicles on campus. This period typically coincides with overlapping lecture schedules, staff working hours, and student movement between classes, resulting in increased demand for parking spaces. In contrast, the lower difficulty reported during the afternoon may be attributed to reduced class density and the departure of early commuters.

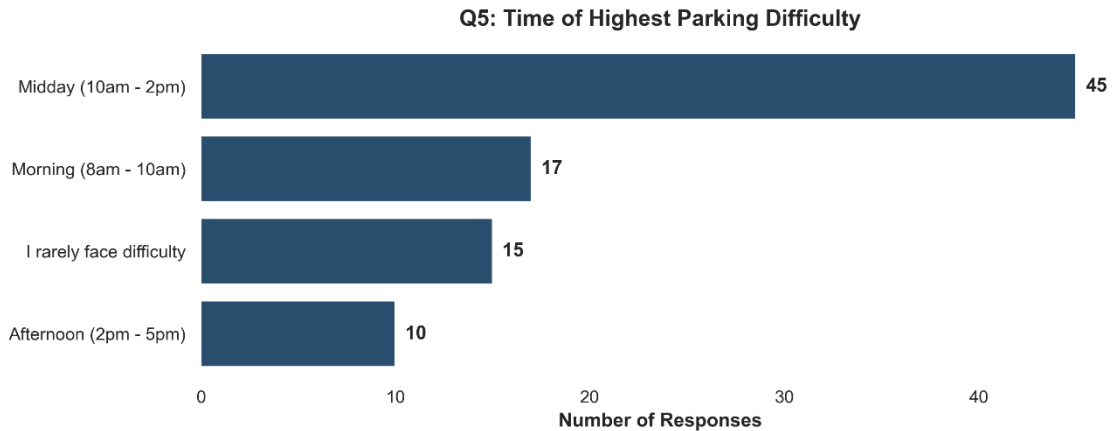


Figure 3.5: Times of Highest Parking Difficulty

f. Question 6: Most Frequently Used Parking Area

Figure 3.6 presents the most frequently used parking areas by respondents. Faculty parking areas recorded the highest usage with 47 responses, indicating that most drivers prefer parking close to academic buildings. This was followed by library parking (13 responses) and STyLE parking (10 responses). Other areas such as DTC/Chancellery parking (8 responses) and hostel parking (6 responses) were less frequently used.

The dominance of faculty parking areas suggests that proximity to lecture halls and workplaces is a primary factor influencing parking choice. Secondary parking locations appear to be used when faculty parking is either unavailable or less convenient, reflecting a hierarchy of parking preferences based on accessibility and walking distance. Since majority of the respondents were from FCI, this suggests that FCI parking is the most frequently used parking areas by students, lecturers and staff.

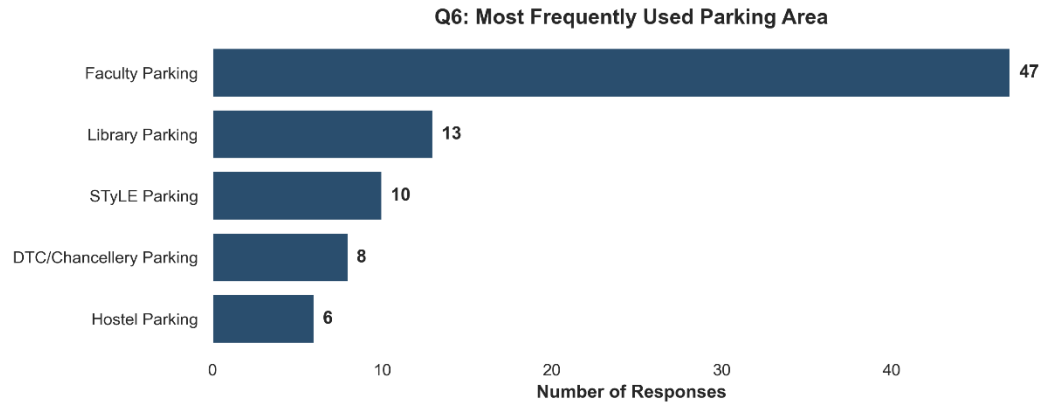


Figure 3.6: Most Frequently Used Parking Area

g. Q7: Area with Most Difficulty Finding Parking

Figure 3.7 highlights the results for **Q7** which indicates that the **Faculty Parking Area** is perceived as the most problematic location for finding available parking, with **44 respondents** identifying it as the area where they experience the greatest difficulty. This value is significantly higher than the other locations, suggesting a clear concentration of parking-related challenges in this zone.

A possible explanation for this trend is the **high daily traffic density** around faculty buildings, where academic staff, students attending lectures, and administrative personnel converge during peak academic hours. Additionally, faculty parking areas are often located close to core facilities, making them a preferred choice due to convenience, which further increases competition for limited parking spaces.

Other areas such as **Library Parking (15 responses)** and **STyLE Parking (13 responses)** show moderate levels of difficulty, likely reflecting periodic congestion during exam periods, events, or peak study hours. In contrast, **Starbeez/Sports Complex Parking**, with only **6 responses**, appears to be the least problematic, possibly due to lower utilization during standard academic hours or better spatial distribution of parking slots.

Overall, the findings highlight that parking difficulty is **not uniformly distributed across the campus** and that targeted interventions; particularly in faculty parking zones would be more effective than campus-wide generalized solutions.

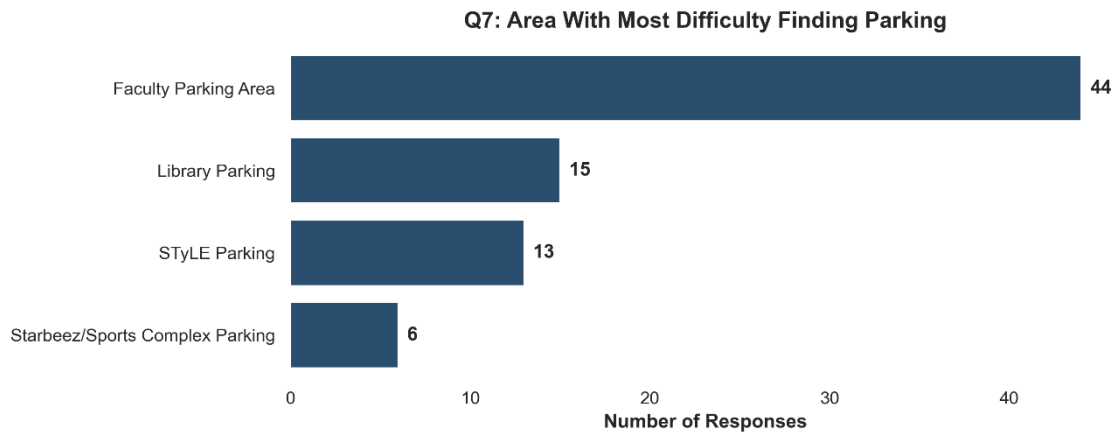


Figure 3.7: Areas where respondents experience the greatest difficulty in finding parking at MMU

h. Q8: Main Parking Issues (Multi-select)

The analysis of Q8 reveals that the most critical parking issues at MMU are related to availability and inefficiency rather than infrastructure alone. The most frequently selected issue is difficulty finding empty parking spaces, reported by 62 respondents, closely followed by time wasted searching for parking, reported by 60 respondents. These two issues together suggest that the core problem is not merely the number of parking spaces, but the lack of visibility and real-time information regarding their availability.

The third most significant issue, lack of information about available slots (37 responses), reinforces this observation. This indicates that users often enter parking areas without knowing whether spaces are available, leading to unnecessary circulation, increased congestion and frustration.

Issues such as congestion during peak hours (34 responses) and poor visibility of parking areas (28 responses) further contribute to the overall parking

experience, particularly during high-demand periods. These factors are likely interconnected, as limited visibility and unclear occupancy status can exacerbate congestion when drivers repeatedly circulate within crowded zones.

Collectively, the results strongly suggest that the primary challenges are operational and informational, rather than purely physical. This directly supports the motivation for introducing a vision-based smart parking system, where real-time occupancy detection and availability display could significantly reduce search time, congestion and user dissatisfaction.

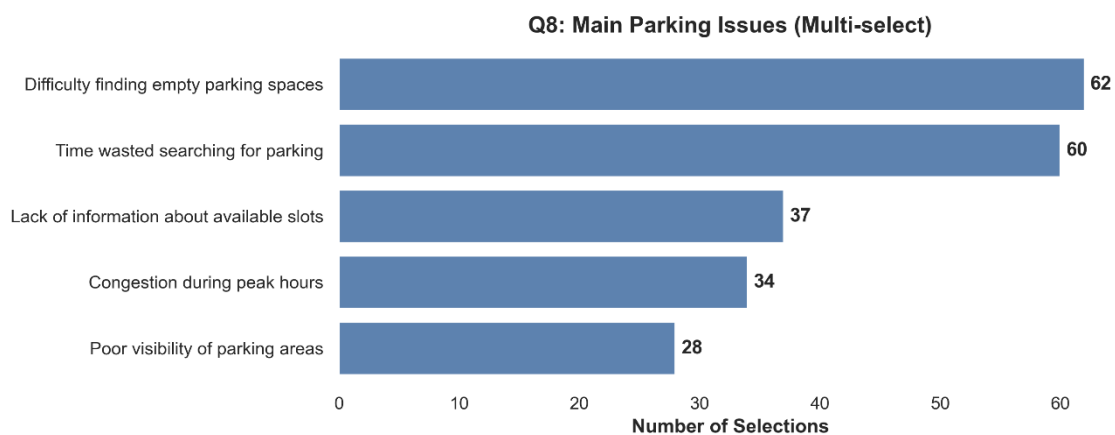


Figure 3.8: Summary of the main parking issues reported by respondents.

i. Q9: Time Spent Searching for Parking

The results for Q9 indicate that a significant proportion of respondents spend a considerable amount of time searching for parking on campus. Most respondents reported spending 5–10 minutes searching for a parking space, followed closely by those who spend 10–15 minutes. A notable group also indicated spending more than 15 minutes, while only a small number of respondents were able to find parking in less than 5 minutes.

This distribution suggests that parking inefficiency is a common experience among MMU users, with most drivers facing moderate to prolonged search times. The relatively low number of respondents who can park quickly implies that available parking spaces are either insufficient or poorly distributed across the campus. These delays may be exacerbated during peak hours, limited visibility of vacant slots, or the absence of real-time parking guidance systems. Overall, the findings highlight a clear need for solutions that can reduce parking search time and improve traffic flow within the campus.

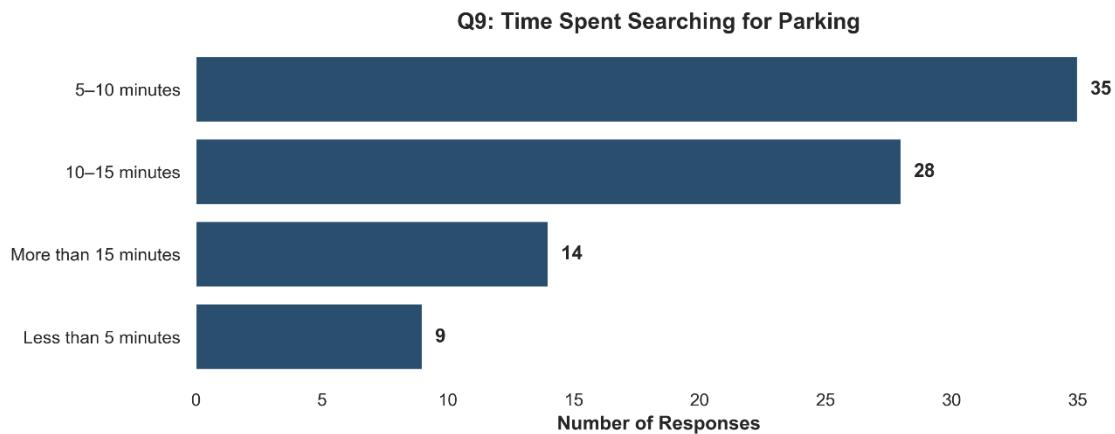


Figure 3.9: Distribution of time spent by respondents searching for parking spaces at MMU.

j. Q10: Satisfaction with Current Parking

The responses to Q10 show that overall satisfaction with the current parking system at MMU is relatively low. The largest group of respondents selected Neutral (41 respondents), indicating uncertainty or mixed experiences regarding the parking situation. However, a substantial number of respondents reported being Dissatisfied or Very dissatisfied (37 respondents), which together outweigh those who expressed satisfaction.

The relatively small number of Satisfied respondents suggests that the existing parking system does not consistently meet user expectations. Neutral responses may reflect adaptation to ongoing parking challenges rather than genuine satisfaction. This trend aligns with earlier findings related to time spent searching for parking and difficulty finding available spaces, reinforcing the conclusion that current parking arrangements lack efficiency and user-friendliness. These results further justify the exploration of intelligent, vision-based parking solutions aimed at improving user satisfaction and overall campus mobility.

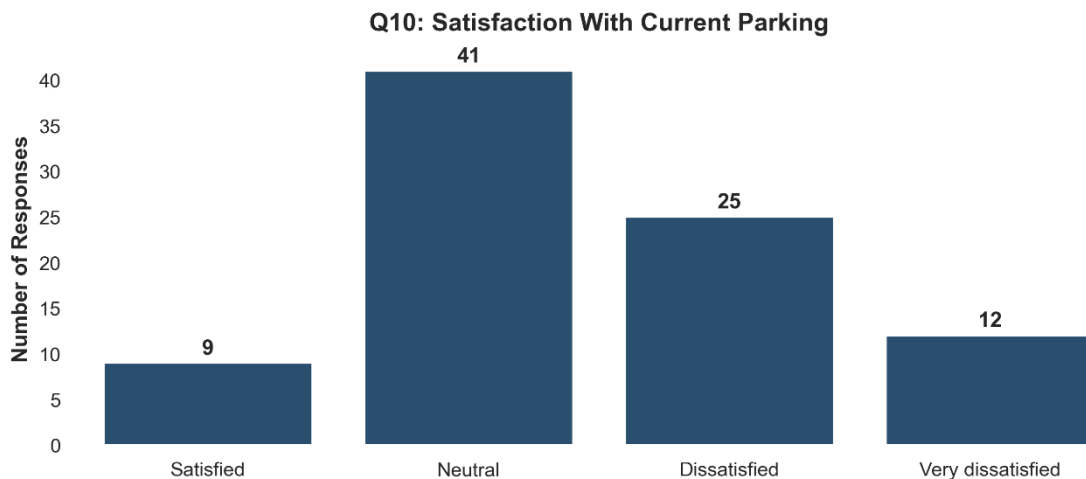


Figure 3.10: Respondents' satisfaction levels with the current parking system at MMU.

k. Q11: Desired Improvements to the Current Parking System

The results for Q11 clearly indicate that respondents prioritize information-driven and efficiency-focused improvements to the current parking system. The most frequently selected improvement is the availability of real-time information on parking availability, highlighting a strong demand for visibility into empty parking spaces before or during the parking search process. This suggests that uncertainty, rather than absolute lack of parking spaces, is a major contributor to parking frustration at MMU.

Closely following this, a significant number of respondents expressed the need for reduced search time, reinforcing the idea that time inefficiency is a critical pain point. Improvements such as better parking layout or guidance and occupied/empty parking signals were also highly selected, indicating that respondents value clear visual cues and structured navigation within parking areas.

Additionally, interest in improved monitoring and management suggests that users perceive the current system as lacking active oversight, especially during peak hours. In contrast, only a very small number of respondents indicated that no improvements are needed, emphasizing a broad consensus that the existing parking experience requires enhancement.

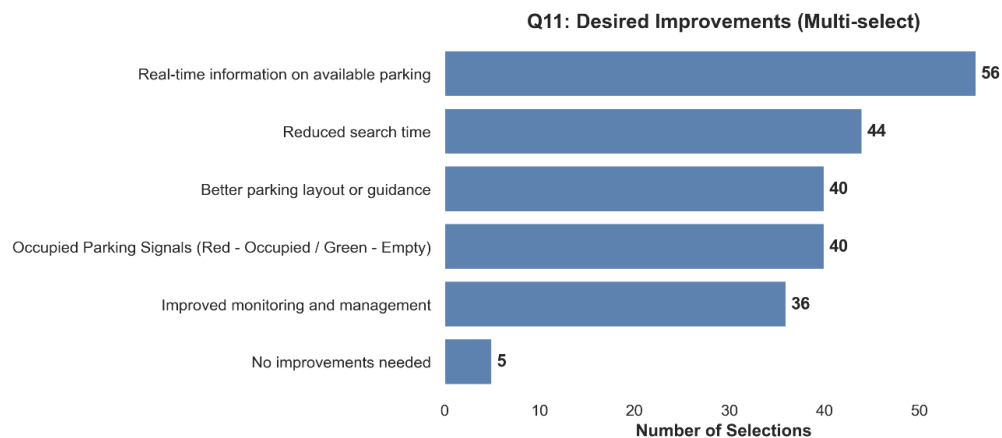


Figure 3.11: Respondents' desired improvements to the current MMU parking system.

I. Q12: Importance of Knowing Parking Availability Before Arrival

The responses to Q12 demonstrate that knowing parking availability in advance is considered highly important by many participants. Most respondents rated this factor as either important or very important, indicating a strong preference for systems that provide pre-arrival or real-time parking information.

This trend reflects the underlying issues identified in earlier questions, particularly time wasted searching for parking and difficulty locating empty spaces. The strong emphasis on availability awareness suggests that users would benefit from predictive or real-time updates that allow them to make informed decisions before entering congested parking areas.

The absence of neutral or negative responses further highlights the perceived value of parking availability information, reinforcing its role as a core functional requirement rather than an optional feature. These findings further validate the motivation behind implementing a vision-based parking monitoring system, as such systems directly address this user need.

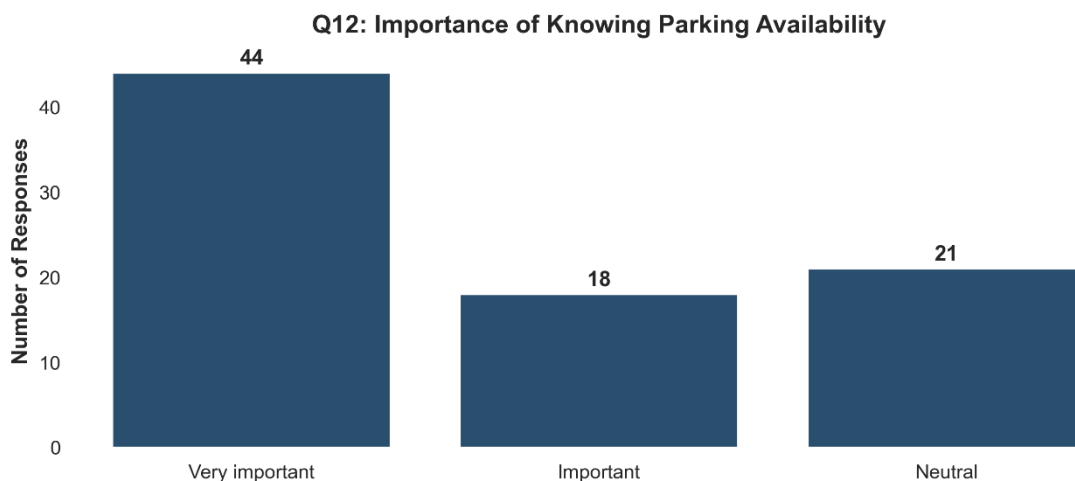


Figure 3.12: Perceived importance of knowing parking availability among respondents.

m. Q13: Essential Features for an Optimal Parking System

The responses to Q13 highlight a strong consensus among participants regarding the core features required for an effective and user-friendly parking system. The most frequently selected feature is a clear visual indication of empty and occupied parking slots, indicating that respondents place high value on immediate and easily interpretable visual cues when navigating parking areas. This reinforces the importance of reducing uncertainty during the parking process.

Closely aligned with this, real-time parking slot availability is also highly prioritized, emphasizing that respondents not only want visual indicators but also expect up-to-date and dynamic information. This suggests that static signage alone is insufficient and that users prefer systems capable of reflecting real-time changes in parking occupancy.

Additionally, a significant number of respondents highlighted the need for an easy-to-understand display or dashboard, which points to usability and simplicity as critical factors for adoption. Features such as accurate and reliable detection further indicate that trust in the system's accuracy is essential, especially if users are expected to rely on it during peak hours. Finally, interest in mobile or web-based viewing reflects a preference for accessibility beyond physical parking locations, allowing users to check availability remotely.

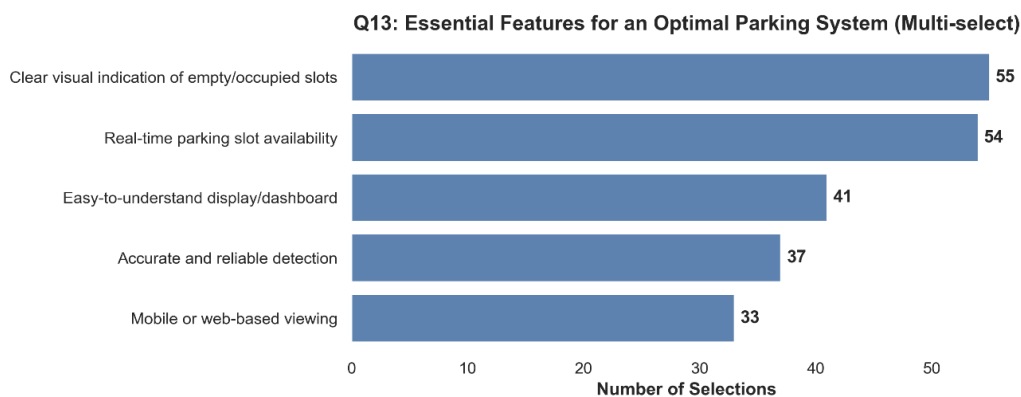


Figure 3.13: Essential features identified by respondents for an optimal parking system

n. Q14: Impact of Parking Availability Display on Reducing Stress

The findings from Q14 show a strong positive perception regarding the impact of parking availability displays on reducing parking-related stress. Almost 70% of respondents either strongly agree or agree that having access to parking availability information would help reduce stress during the parking process. This indicates that uncertainty and lack of information are significant contributors to negative parking experiences.

The presence of a moderate number of neutral responses suggests that while most users anticipate benefits, some may require direct experience with such systems before fully recognizing their impact. Importantly, very few respondents disagreed with the statement, suggesting minimal resistance to the concept of availability displays.

These results support the argument that providing real-time parking information can improve not only efficiency but also the overall user experience and emotional response to parking.

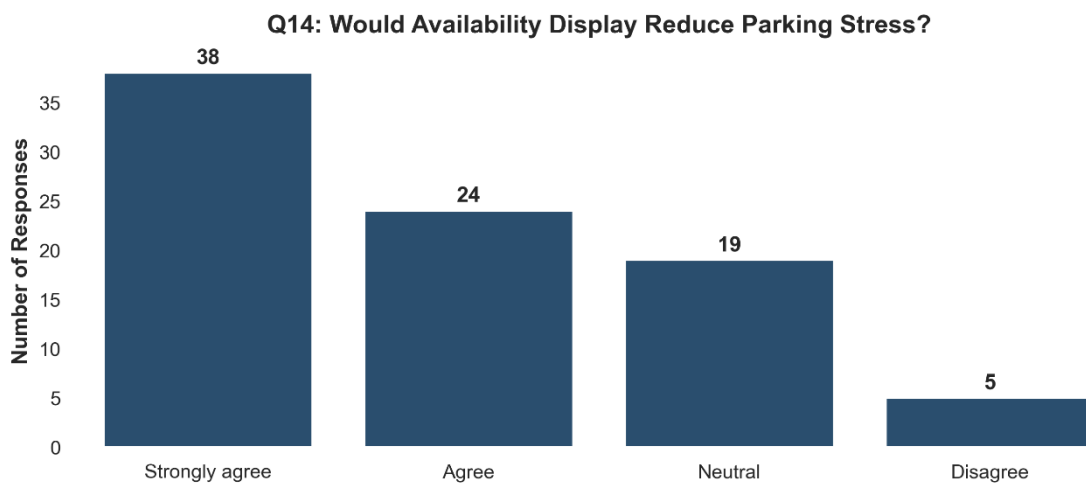


Figure 3.14: Respondents' perceptions of whether parking availability displays can reduce parking stress.

o. Q15: Previous Use of Parking Applications

The results indicate that a slight majority of respondents (55%) have not used a parking application before, while 45% reported prior experience with such applications. This suggests that although parking-related applications are not yet widely adopted among MMU users, there is still a substantial portion of the population familiar with digital parking solutions. The limited prior exposure highlights the importance of designing an intuitive and easy-to-use system to encourage adoption among first-time users.

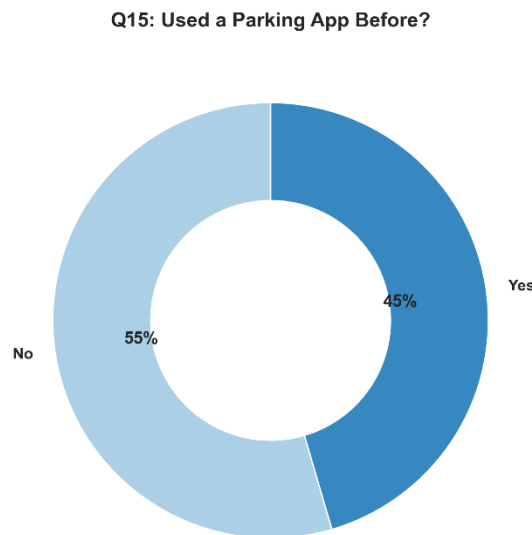


Figure 3.15: Distribution of respondents based on prior usage of parking applications.

p. Q16: Parking Application Features Used (Multi-select)

Among respondents who have used parking applications, the most commonly utilized feature is paying for parking, followed by viewing parking rates or fees. Features related to parking duration, tracking and checking parking availability were moderately selected. While reserving a parking space and viewing parking rules or restrictions received fewer responses. These results suggest that users primarily rely on parking applications for transactional and cost-related purposes,

while real-time availability and reservation features are less frequently used; likely due to limited availability of such functionalities in existing systems.

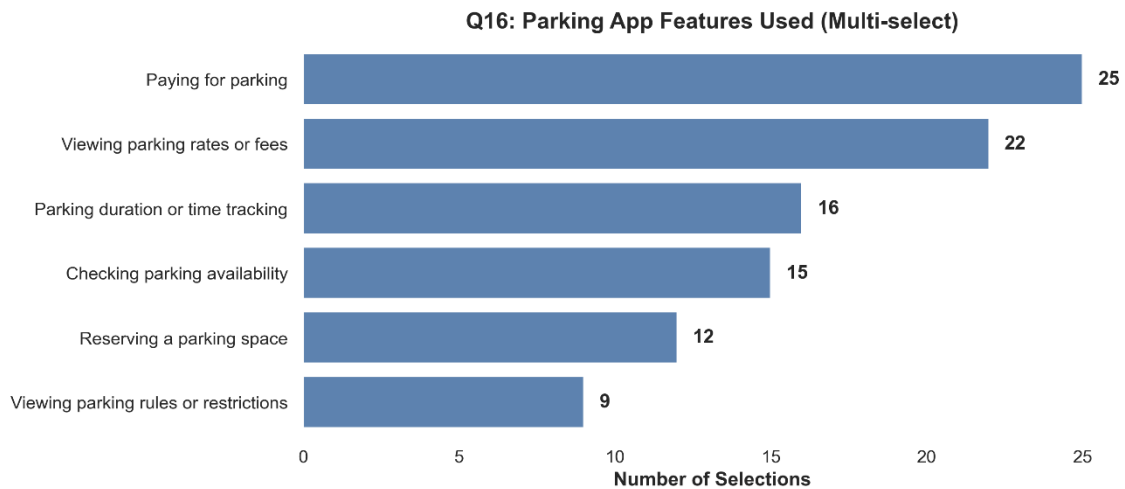


Figure 3.16: Frequency of parking application features used by respondents.

q. Q17: Perceived Usefulness of a Vision-Based Parking System

Most respondents showed a positive perception of a vision-based parking system. An equal number of participants rated the system as “very useful” or “useful”, together forming a clear majority. A smaller portion of respondents remained neutral, while only a limited number perceived the system as not useful.

This strong positive inclination indicates high user acceptance and reinforces the relevance of implementing computer vision-based parking solutions within the MMU campus.

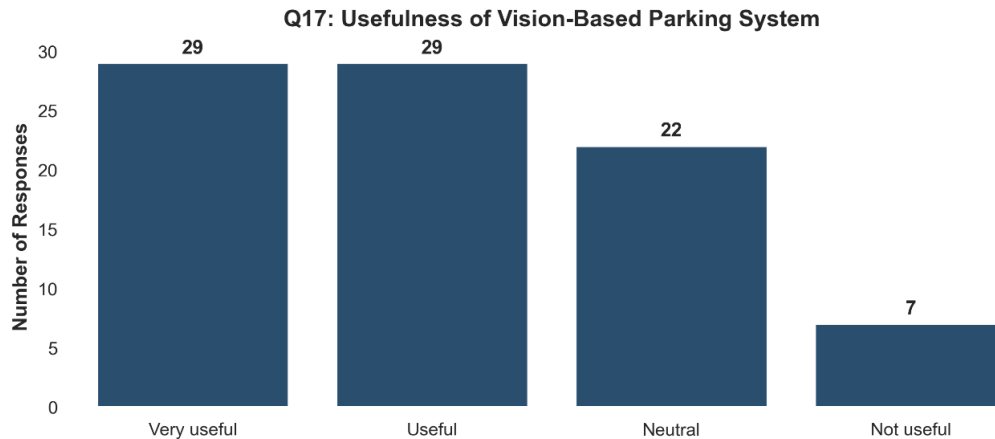


Figure 3.17: Respondents perceived usefulness of a vision-based parking system.

r. Q18: Support for Implementing the System at MMU

An overwhelming majority of respondents (64 participants) indicated clear support for implementing a smart parking system at MMU. A smaller group selected “maybe,” reflecting cautious optimism, while only a minimal number opposed the implementation. This strong level of support demonstrates readiness among the campus community for adopting a technology-driven parking solution and provides strong justification for further development and deployment.

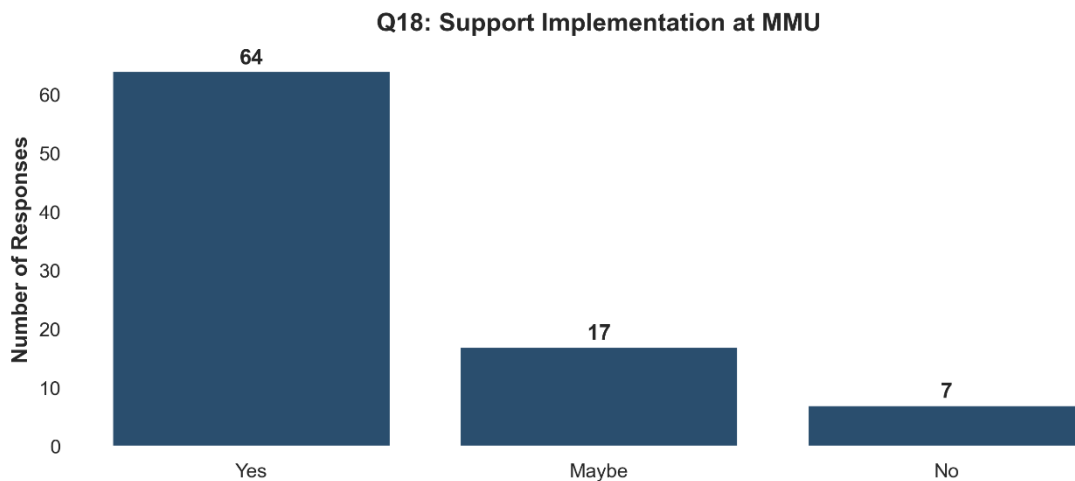


Figure 3.18: Level of respondent support for implementing a smart parking system at MMU.

In summary, the survey analysis highlights the significance of parking-related challenges faced by users and reinforces the need for an improved parking monitoring solution. The results show that a large portion of respondents are either vehicle owners or are regularly affected by parking congestion, indicating that parking availability is a common and recurring issue. Participants expressed clear frustration with the difficulty of locating available parking spaces, particularly during peak hours, which negatively impacts their overall campus experience.

Furthermore, the findings suggest a generally positive attitude toward the adoption of a real-time parking slot monitoring system, provided that the system is accurate, reliable, and easy to use. The emphasis on simplicity reflects users' preference for a solution that requires minimal learning effort while delivering clear and timely information.

Overall, the survey results provide strong justification for the proposed system and serve as a practical basis for defining system requirements in the subsequent sections.

Key takeaways from the survey analysis include:

- A majority of respondents are **directly affected by parking availability and congestion**, either as vehicle owners or frequent users of campus parking.
- Parking difficulties are perceived as a **significant problem**, contributing to wasted time and increased frustration.
- Respondents demonstrate a **positive perception toward a real-time parking monitoring system**, indicating readiness for technological intervention.
- **Ease of use and system simplicity** are critical factors influencing user acceptance and adoption.

- The survey findings strongly support the need for an **efficient, user-friendly, and reliable parking management solution**.

While the survey analysis provides valuable insights into users' perceptions and experiences regarding parking availability, these findings are primarily based on subjective responses. To strengthen the reliability of the results, it is necessary to examine whether these perceptions are reflected in actual parking conditions.

Therefore, the following section presents an **observation-based study**, focusing on real-world parking usage and behavior within the selected area. By directly observing parking occupancy patterns and user behavior, this section aims to validate the survey outcomes and provide concrete evidence that supports the identified issues.

The combination of survey and observation results enables a more comprehensive understanding of the problem and forms a stronger basis for subsequent requirement analysis and system design.

3.2.3 Observation

Based on the findings obtained from the survey analysis in the previous section, an observation-based study was conducted to further examine parking usage patterns and validate the issues reported by respondents. The survey results indicated that several parking areas experience higher demand and greater levels of congestion, particularly during peak hours. As a result, three parking locations were selected as the primary observation sites for this study: **FCI Parking** and **FAIE Parking**. These locations were identified as the most frequently used parking areas and were commonly reported by respondents as areas where parking difficulties are most prominent.

The observation process focused on monitoring real-world parking conditions at different times of the day in order to capture variations in occupancy

and user behavior. Observations were conducted during multiple time periods, including peak and non-peak hours, to ensure that the collected data reflects typical parking usage patterns. Key aspects observed include the availability of parking spaces, vehicle circulation behavior and general congestion levels within each parking area.

To support the observational findings, photographic evidence was planned to be collected for each selected parking location. These images are intended to visually demonstrate parking occupancy conditions at different times of the day and provide supporting evidence for the analysis presented in the following subsection. The images will be added after data collection is completed.

a. FCI Parking Observation

FCI Parking was selected due to its high usage by students and staff, as reflected in the survey responses. This parking area frequently experiences congestion, particularly during lecture hours and peak academic periods.



Figure 3.19: FCI Parking (8 - 10 am)



Figure 3.20: FCI Parking (10 - 2 pm)



Figure 3.21: FCI Parking (2 - 5 pm)

b. FAIE Parking Observation

FAIE Parking was identified through the survey as another high-demand parking area. Respondents highlighted recurring difficulties in finding available spaces, especially during busy academic schedules.



Figure 3.22: FAIE Parking (8 - 10 am)



Figure 3.23: FAIE Parking (10 - 2 pm)



Figure 3.24: FAIE Parking (2 - 5 pm)

The observations collected from these two parking locations provide real-world context to the issues identified in the survey analysis. By examining parking conditions across multiple locations and time periods. This study ensures that the findings are grounded in actual usage patterns. The detailed analysis of these observations is presented in the next subsection to further validate and support the survey results.

3.2.3.1 Analysis on results

The observation results provide practical evidence that supports and reinforces the findings obtained from the survey analysis. By examining real-world parking conditions at FCI Parking and FAIE Parking. Clear correlations can be established between users' reported experiences and the actual parking usage patterns observed on-site.

During the **morning period (8:00 a.m. to 10:00 a.m.)**, the observed parking space across all two locations ranged from **half-full to nearly full**. This observation corresponds closely with survey responses to questions related to parking usage frequency and time-of-day congestion. Where respondents indicated that parking availability begins to decline shortly after the start of academic activities. Although parking spaces remain accessible during this period. The increasing occupancy supports respondents' perception that early congestion gradually develops as the day progresses.

Observations made during the **midday period (10:00 a.m. to 2:00 p.m.)** revealed that parking areas were frequently **jammed or overfilled**. Particularly in high-demand zones such as FCI Parking. This finding strongly correlates with survey questions that asked respondents to identify the most challenging time for parking, where almost 50% of participants highlighted midday as the period with the highest difficulty. The observed lack of available parking spaces during this time further validates survey responses related to increased frustration, time wastage and the need for improved parking visibility.

In the **late afternoon and evening period (from 2:00: p.m. to 5:00 p.m.)** observations showed a noticeable reduction in parking spaces with many spaces becoming **partially empty**. This trend aligns with survey responses indicating fewer parking-related issues outside peak academic hours. Respondents who reported lower difficulty during non-peak periods are supported by the observed decrease in congestion as academic and administrative activities conclude for the day.

The patterns observed across the selected parking locations demonstrate a strong alignment with multiple survey questions addressing parking difficulty, time-based congestion and user experience. The consistency between user-reported data and direct observations indicates that the parking issues identified in the survey are not isolated perceptions but are grounded in actual campus conditions. This strong correlation between survey and observation findings reinforces the reliability of the data and highlights the importance of implementing a real-time parking monitoring system to address peak-hour congestion and improve overall parking efficiency.

3.3 Requirement

This section presents the requirements analysis for the proposed parking monitoring system, derived directly from the findings of the survey and observation studies discussed in the previous sections. The purpose of this section is to translate identified user needs, challenges and real-world parking conditions into well-defined system requirements. By doing so, the proposed system is designed based on evidence rather than assumptions, ensuring that it effectively addresses the parking issues experienced by users.

The requirements outlined in this section serve as a foundation for the system design and implementation phases. They define both the core functionalities the system must support and the quality attributes it must maintain

in order to operate reliably in real-world environments. The requirements are categorized into functional, non-functional and user requirements to provide a clear and structured representation of system expectations.

3.3.1 Functional Requirements

Functional requirements describe the essential operations and features that the system must perform to meet user needs and resolve the parking issues identified through the survey and observation analyses. These requirements focus on system behavior, data processing and user interaction with the parking monitoring solution.

a. FR-1: Parking Slot Occupancy Detection

The system shall be able to detect and determine the occupancy status of individual parking slots in real time. This requirement is derived from survey findings where users reported difficulty in identifying available parking spaces, particularly during peak hours and is further supported by observation results showing high congestion during midday periods.

b. FR-2: Video Feed Input Handling

The system should accept video feeds as input from cameras installed at monitored parking locations or the recordings obtained from the devices used. This requirement ensures that the system can continuously receive visual data required for analyzing parking slot occupancy.

c. FR-3: Frame Extraction from Video Feed

The system should extract individual frames from the video feed for further processing. This requirement supports the use of computer vision techniques by enabling frame-by-frame analysis of parking areas.

d. FR-4: Parking Slot Status Classification

The system shall classify each detected parking slot as either occupied or empty based on the output of the occupancy detection model. This requirement enables accurate identification of parking availability at the slot level.

e. FR-5: Real-Time Parking Availability Update

The system should update parking slot occupancy status in real time or near real time as new detection results become available. This requirement ensures that users receive timely and accurate parking information, addressing variations in parking availability observed throughout the day.

f. FR-6: Visual Representation of Parking Slot Status

The system should provide a clear visual representation of parking slot occupancy using colour indicators to distinguish between occupied and empty slots. This requirement aligns with survey results where respondents expressed a preference for an intuitive and easily understandable method of viewing parking status.

g. FR-7: Support for Multiple Parking Locations

The system shall support monitoring of multiple parking locations, specifically FCI Parking and FAIE Parking. This requirement is based on survey and observation findings indicating that parking congestion occurs in more than one high-demand area on campus.

h. FR-8: Parking Location Selection Interface

The system shall allow users to select a parking location to view through a simple interface element such as tabs or buttons. This requirement supports ease of navigation and enables users to switch between parking locations efficiently.

i. FR-9: Display of Parking Layout

The system shall display the parking layout corresponding to the selected parking location, reflecting the physical arrangement of parking slots. This requirement improves user understanding by presenting parking availability in a familiar spatial format.

j. FR-10: Summary of Parking Availability

The system shall display a summary showing the total number of available and occupied parking slots for the selected parking location. This requirement allows users to quickly assess overall parking conditions without analyzing individual slots.

3.3.2 Non-functional Requirements

Non-functional requirements define the quality attributes and constraints of the proposed parking monitoring system. Unlike functional requirements, which focus on what the system does, non-functional requirements describe how well the system should perform under real-world conditions. These requirements are essential to ensure that the system remains reliable, usable and effective. Particularly during peak parking periods identified in the survey and observation analyses.

The non-functional requirements presented in this section are derived from user expectations expressed in the survey as well as practical considerations observed during on-site parking analysis.

a. NFR-1: System Accuracy

The system should provide accurate parking occupancy information to users. Accuracy is a critical requirement, as survey respondents highlighted frustration caused by uncertainty when searching for available parking spaces. Observations showing high congestion during peak hours further emphasize the need for reliable detection to prevent misinformation.

b. NFR-2: System Response Time

The system should update and display parking availability information within a reasonable time frame. Timely updates are important to ensure that users receive current information, especially during midday peak periods where parking conditions change rapidly.

c. NFR-3: Usability and Ease of Use

The system should be simple and intuitive to use, requiring minimal effort from users to understand parking availability. This requirement is directly supported by survey responses where users expressed a preference for a clear and straightforward interface without unnecessary complexity.

d. NFR-4: Scalability

The system should be designed in a way that allows future expansion, such as the inclusion of additional parking locations if required. While the current scope focuses on selected parking areas, scalability ensures that the system remains adaptable without requiring major redesign.

3.3.3 User Requirements

User requirements describe the expectations and needs of end users when interacting with the proposed parking monitoring system. These requirements are expressed from the user's perspective and focus on usability, clarity and practicality rather than technical implementation details. The user requirements presented in this section are derived primarily from the survey findings and are further supported by real-world observations of parking behavior.

a. UR-1: Access to Clear Parking Availability Information

Users should be able to easily view the availability status of parking spaces within the monitored areas. This requirement is based on survey responses where users expressed difficulty in identifying available parking slots and a strong need for clearer parking information, particularly during peak hours.

b. UR-2: Simple and Intuitive Interface

Users should be able to understand and use the system without requiring prior technical knowledge or extensive guidance. This requirement reflects survey feedback indicating that ease of use is a key factor influencing user acceptance of a parking monitoring solution.

c. UR-3: Minimal Time and Effort to Obtain Information

Users should be able to access parking availability information quickly, without unnecessary steps. This requirement is supported by both survey and observation findings, where users reported frustration due to time wasted searching for parking spaces.

d. UR-4: Confidence in Displayed Information

Users should be able to trust the accuracy of the information provided by the system. This requirement aligns with survey responses emphasizing the importance of reliable information to support decision-making when selecting parking areas.

In conclusion, this section has presented a structured requirements analysis for the proposed parking monitoring system, derived from both survey findings and observation results. By translating user feedback and real-world parking behavior into functional, non-functional and user requirements, the system requirements are clearly defined and aligned with actual needs.

Moreover, these requirements provide a solid foundation for the subsequent system design phase, ensuring that all design decisions are guided by validated evidence rather than assumptions. The requirements outlined in this section will

be referenced throughout the system design and implementation stages to maintain consistency and traceability.

Chapter 4: System Design

4.1 Overview

This chapter presents the system design for the proposed parking monitoring system. The purpose of this chapter is to illustrate how the system requirements identified in Chapter 3 are translated into a structured and practical system design. The system design provides a high-level view of how different system components interact, as well as a detailed representation of system behavior and user interaction.

The designs presented in this chapter focus on clarity and simplicity to ensure that the proposed solution is feasible within the project scope. Various diagrams are used to visually represent the system structure, processes, and interactions, enabling a clearer understanding of how the system operates as a whole. These diagrams also serve as a reference for the implementation phase, ensuring consistency between system requirements and system development.

4.2 Rich Picture Diagram

Figure 4.1 presents the Rich Picture Diagram for the proposed parking monitoring system. The diagram provides a high-level, non-technical illustration of the current parking situation within the MMU campus and highlights the key issues experienced by users. It visually represents the interaction between drivers, the parking environment, and the proposed system solution.

As illustrated in the diagram, drivers frequently encounter congested parking areas, resulting in repeated circulation, time wastage, and frustration due to the lack of real-time parking availability information. These issues are consistent with the findings from the survey and observation analyses discussed in Chapter 3, where users reported difficulty in identifying available parking spaces during peak hours.

The diagram also introduces the proposed solution concept, which involves the use of camera-based monitoring, a processing component, and a user-facing display to provide parking availability information. This conceptual representation demonstrates how real-time information can assist drivers in making informed parking decisions, thereby reducing congestion and unnecessary circulation within parking areas.

To conclude with, the Rich Picture Diagram serves to contextualize the parking problem and proposed solution from a real-world perspective. It establishes a clear link between the identified problems and the motivation for the system design, forming a foundation for the more detailed technical designs presented in the subsequent section

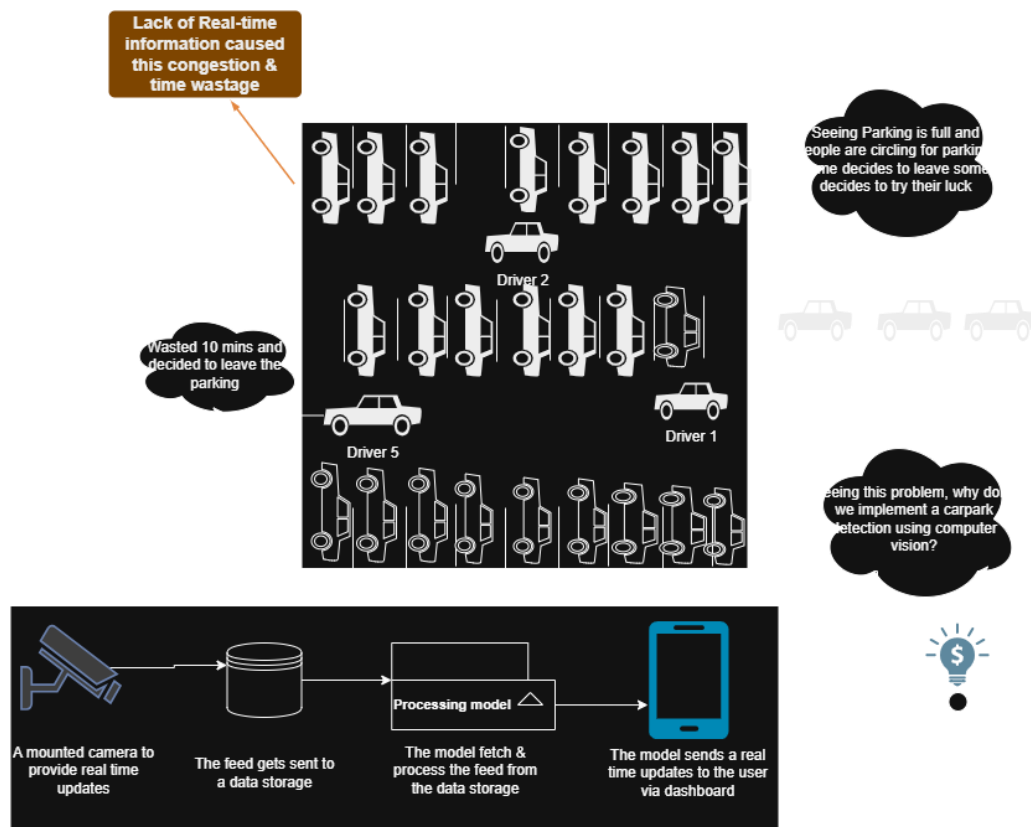


Figure 4.1: Rich Picture Diagram

4.3 Use Case Diagram

The Use Case Diagram illustrates how the viewer interacts with the parking monitoring prototype system. In line with the project scope, the system is implemented as a dashboard-based prototype where interaction is kept simple and focused on viewing parking information rather than application-level features. As shown in the diagram, the viewer can select between two monitored parking locations, FCI and FAIE, using tabs or buttons on the dashboard. Once a location is selected, the system displays the corresponding parking layout, including slot availability (occupied/empty) and a summary count of available and occupied slots. The video feeds for both locations are treated as input sources to the prototype system, which processes the footage and updates the dashboard accordingly.

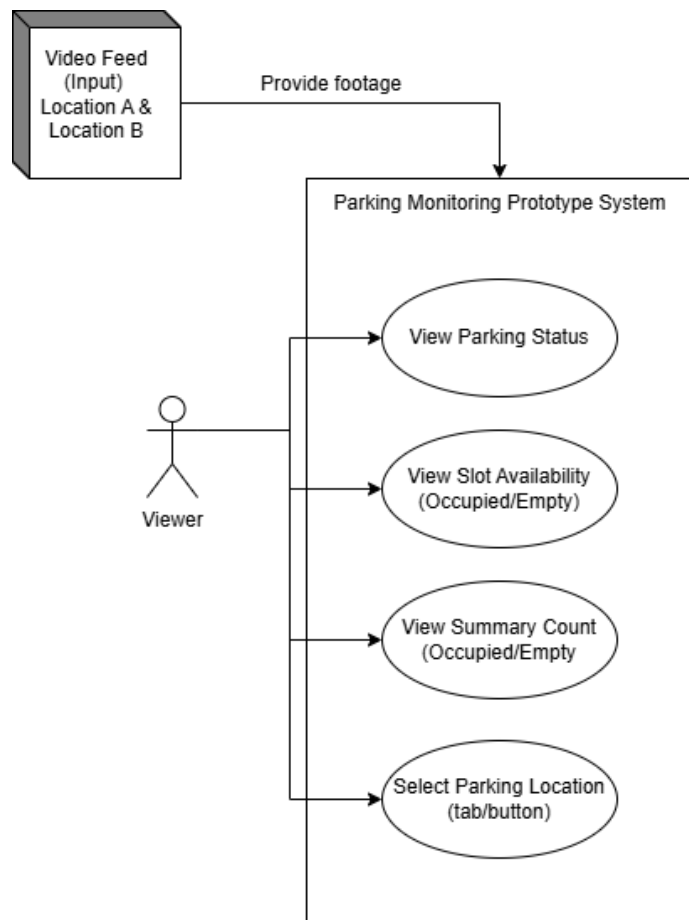


Figure 4.2: Use Case Diagram

4.4 Activity Diagram

Figure 4.3 illustrates the activity flow of the proposed parking monitoring prototype. The process begins when the system receives a video feed from the selected parking area (FCI & FAIE) and extracts frames for analysis. Each frame is processed by the prototype system, where the parking detection model determines the occupancy status of parking slots. Based on the detection outcome, the system updates the slot status and recalculates the overall availability summary. The dashboard is then refreshed to display the latest parking status to the viewer. This workflow repeats continuously as new frames are processed, allowing the dashboard to reflect real-time changes in parking occupancy. In the dashboard interface the viewer can switch between **FCI** and **FAIE** using tabs or buttons. The displayed parking layout is refreshed based on the selected location's latest processed results.

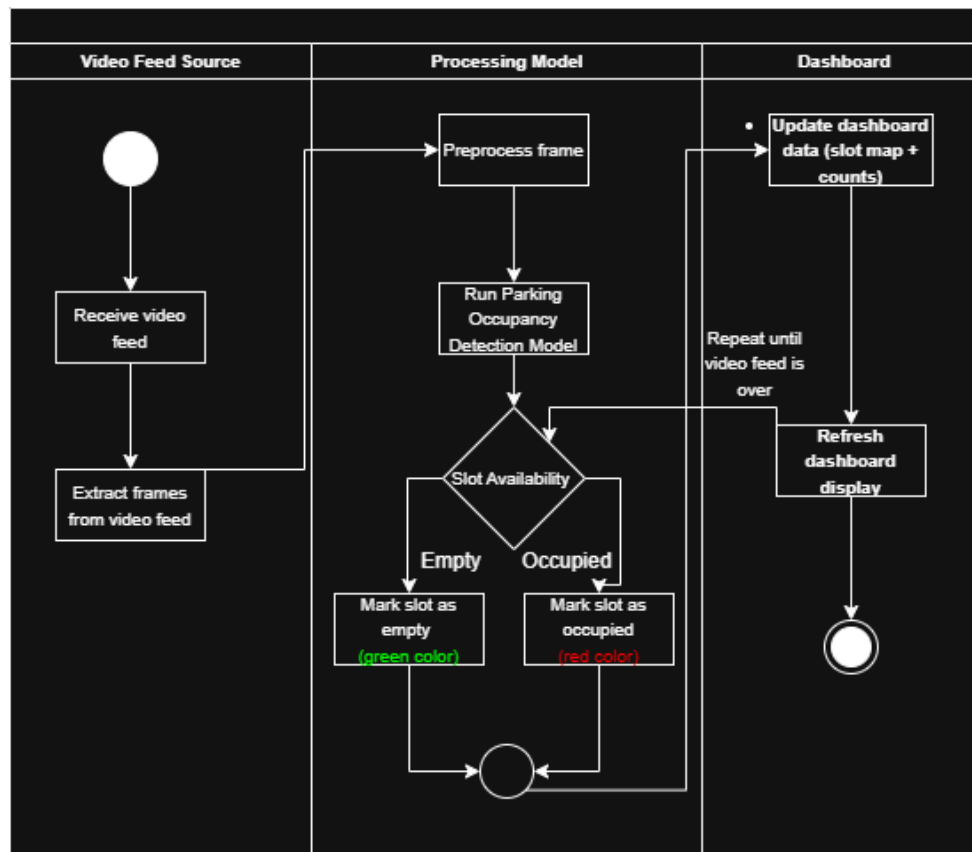


Figure 4.3: Activity Diagram

4.5 Class Diagram

Figure 4.4 presents the UML class diagram of the proposed parking monitoring system designed to operate across two parking locations **FCI** and **FAIE**. The class diagram presents the core data entities and their relationships, focusing on how parking occupancy information is derived from video input and displayed on the dashboard. The design intentionally remains lightweight to reflect the prototype-based nature of the system.

The VideoFeed class represents the source of parking footage for a specific parking location. Each video feed is associated with a single ParkingLocation and produces multiple Frame objects that are extracted for analysis. These frames are processed by the OccupancyDetector, which encapsulates the parking occupancy detection model used in the prototype. The detector analyses each frame and generates a set of DetectionResult objects containing the predicted occupancy status for individual parking slots.

Parking spaces are represented using the ParkingSlot class. Which stores essential information such as slot identity, region of interest, current occupancy status and last update time. Each parking slot belongs to a specific ParkingLocation, allowing the system to differentiate between slots from different monitored areas. The occupancy state of each slot is standardised using the SlotStatus enumeration, which defines the possible states as **occupied** or **empty**.

The Dashboard class represents the user-facing component of the prototype. It maintains a collection of available parking locations and allows the viewer to select between **FCI** and **FAIE** using tabs or buttons. Based on the selected location, the dashboard displays the corresponding parking layout, slot availability, and summary counts of occupied and empty slots. This structure supports multi-location monitoring while keeping user interaction minimal and consistent with the project scope.

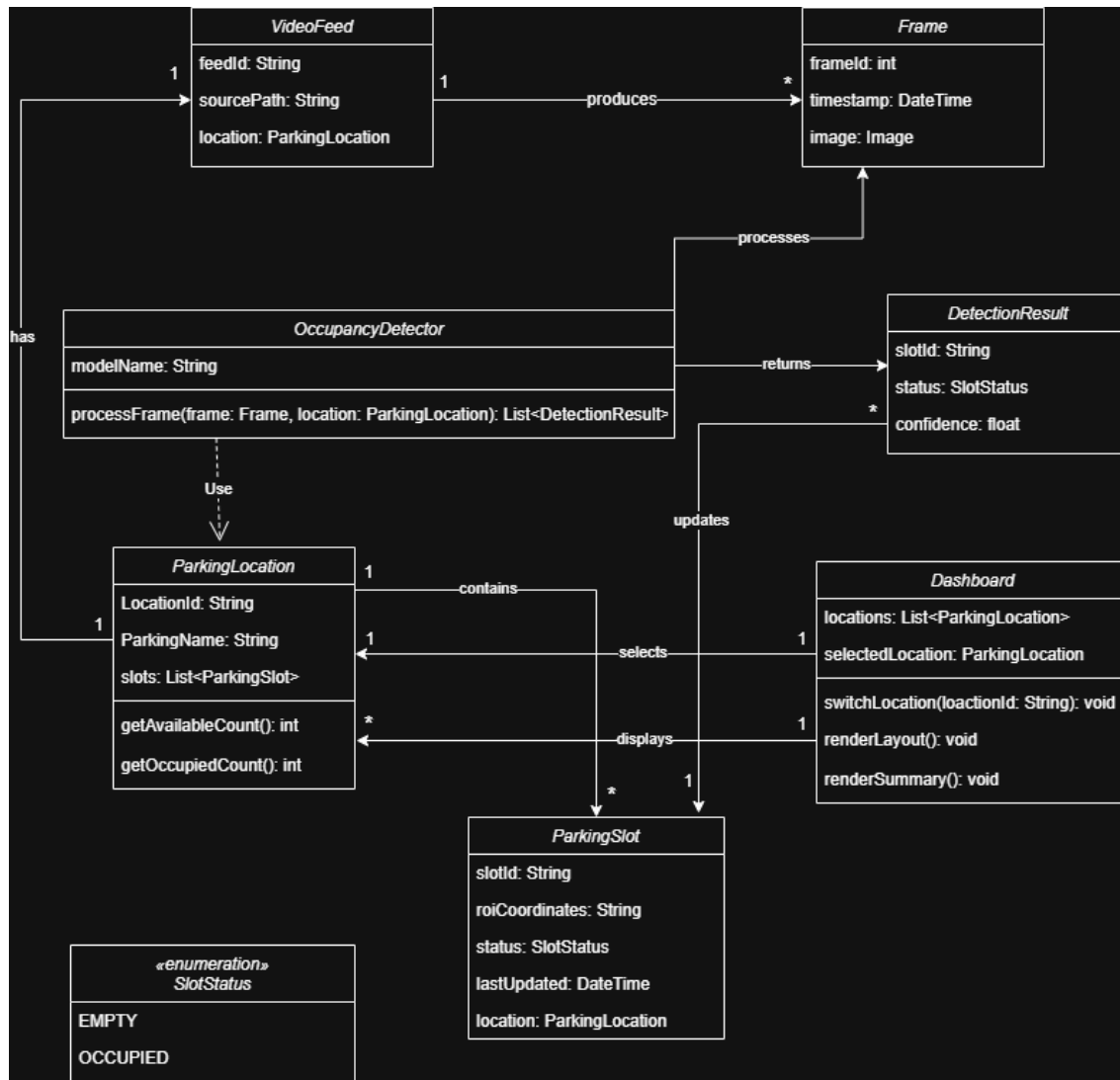


Figure 4.4: UML Class Diagram

4.6 Sequence Diagram

Figure 4.5 shows the sequence of interactions within the parking monitoring prototype system. Focusing on how parking occupancy information is generated and displayed when the viewer switches between **FCI** and **FAIE**. The sequence begins when the viewer selects a parking location using the dashboard tabs/buttons. Once a location is selected the dashboard retrieves the corresponding video feed frames and forwards them to the occupancy detection

module for processing. The detector preprocesses the frames, performs occupancy detection and returns detection results indicating whether each parking slot is occupied or empty.

The dashboard then updates the slot status information and refreshes the displayed parking layout and summary counts for the selected location. This interaction sequence repeats as new frames are processed, allowing the dashboard to reflect changes in parking occupancy over time, while maintaining a simple viewer-based interaction consistent with the prototype scope.

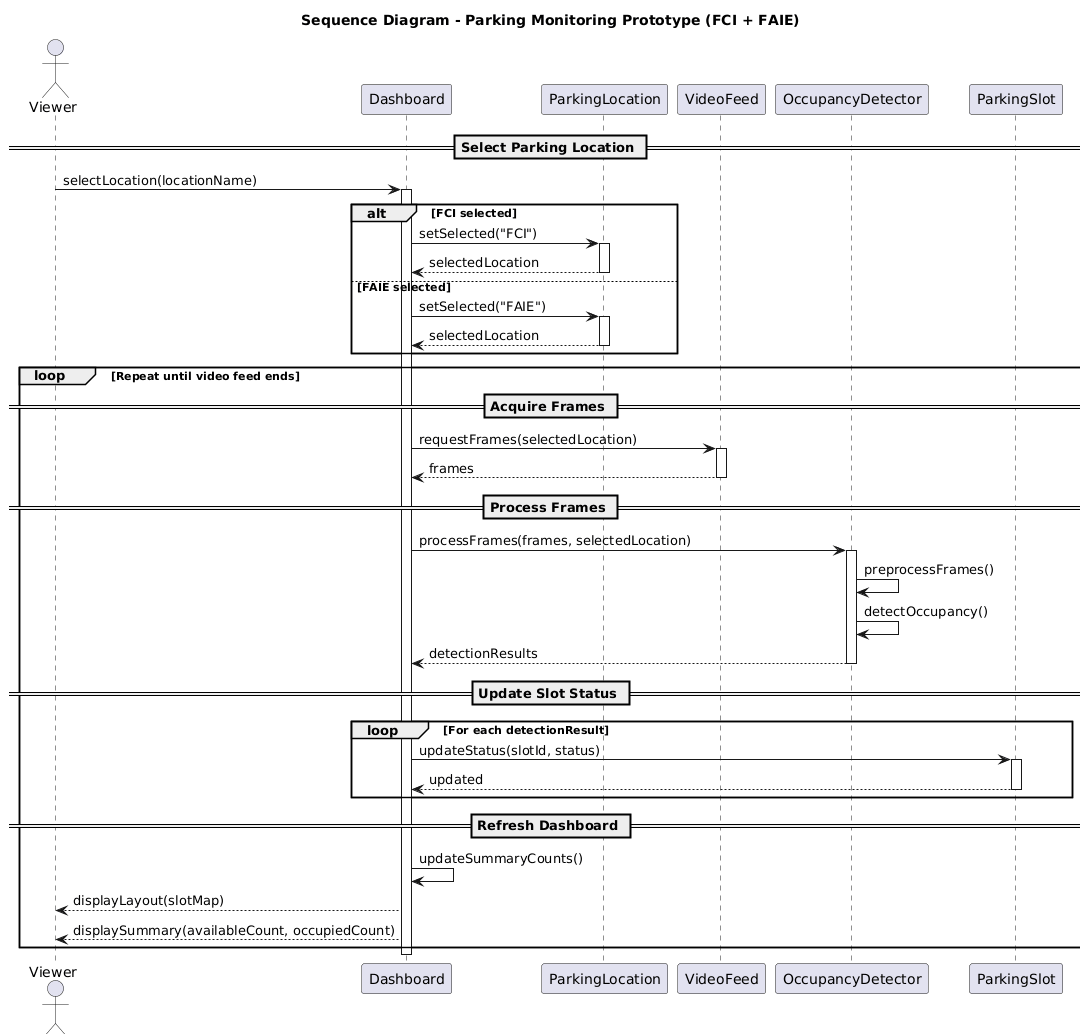


Figure 4.5: Sequence Diagram

4.7 Interface Design

Figure 4.6 gives an idea of the home interface of the parking monitoring system. This screen allows the user to select which parking location to view by choosing between **FCI** and **FAIE**. The interface is intentionally minimal to reduce user effort and ensure that navigation remains simple. This design aligns with the prototype scope of the project, where the primary goal is to provide quick access to parking availability information rather than full application functionality.

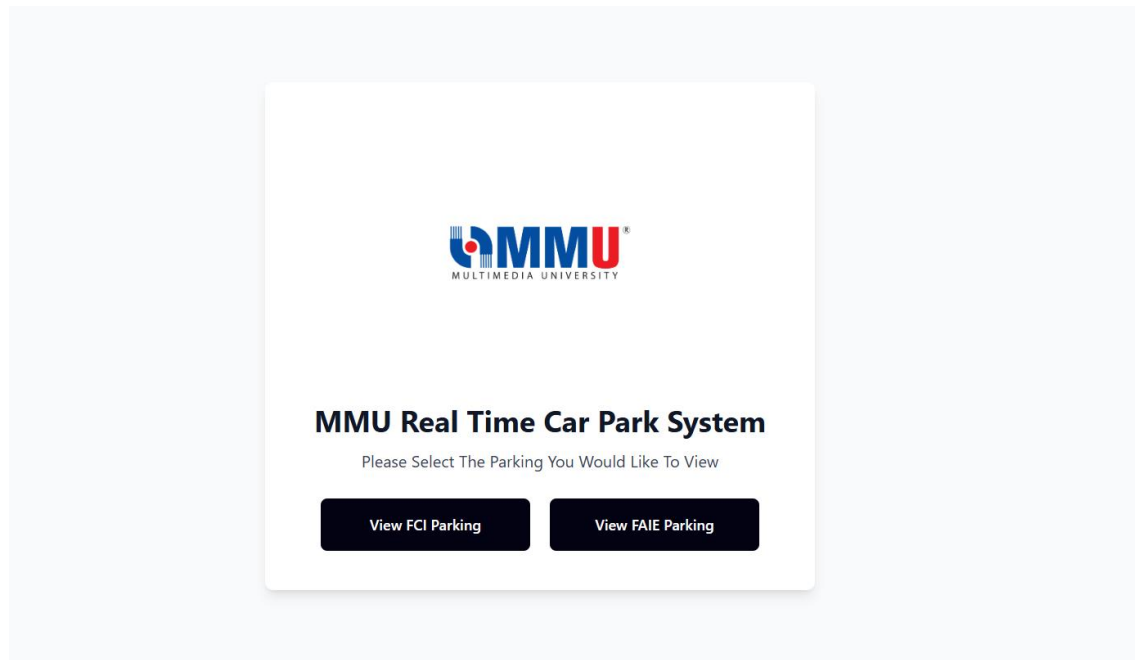


Figure 4.6: Home Page Interface

Figure 4.7 presents the dashboard interface for the **FCI parking area**, which features an angled parking layout. The interface reflects the actual physical configuration of the parking area by displaying angled slots on both sides of the main drive lane. Each slot is uniquely labelled and colour-coded to represent its occupancy status. This design approach enhances usability by allowing users to intuitively interpret parking availability based on a layout that closely resembles the real parking environment.

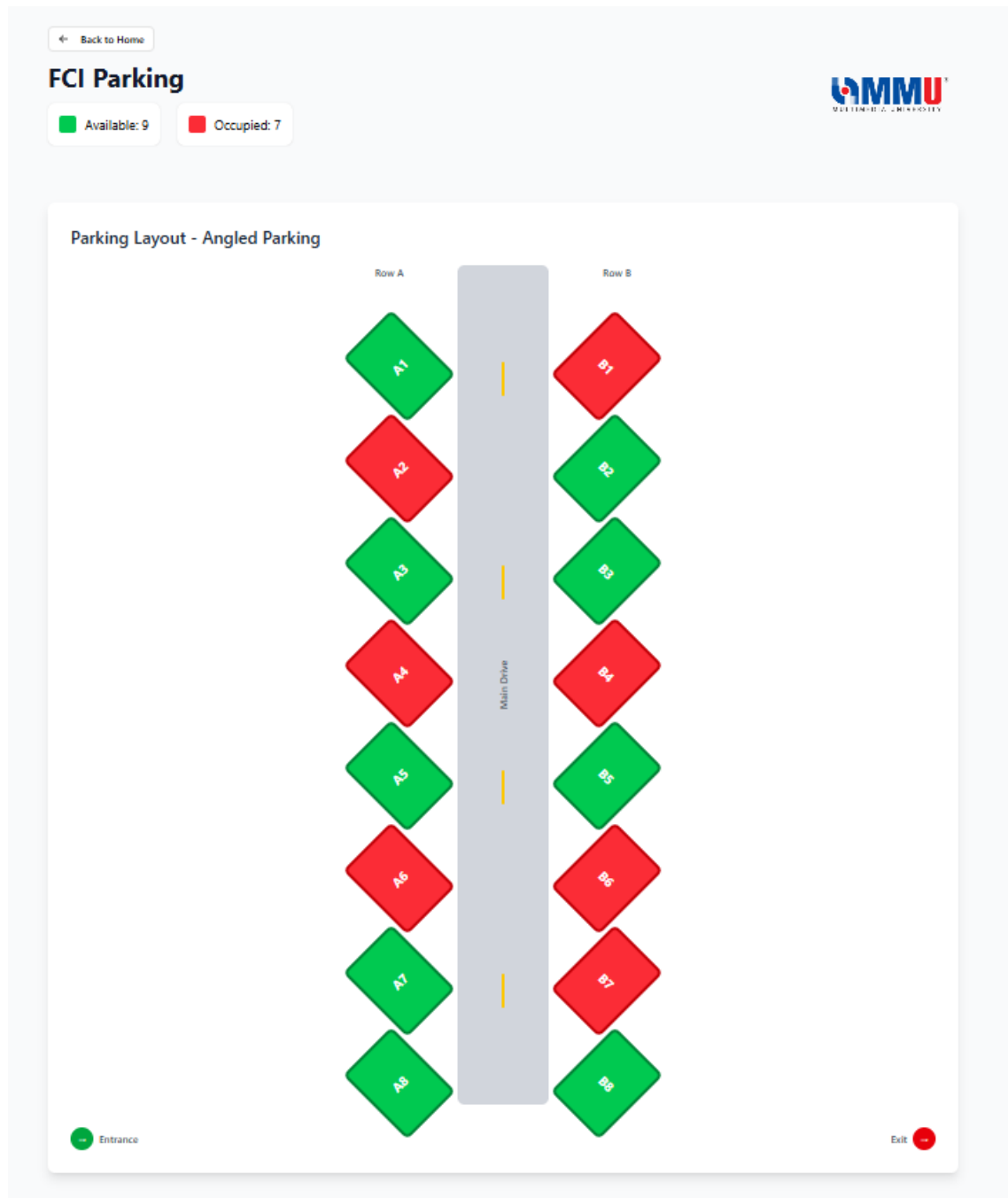


Figure 4.7: FCI Parking Interface

Figure 4.8 illustrates the dashboard interface for the **FAIE parking area**, which follows a perpendicular parking layout. Parking slots are organised into rows and displayed according to their physical arrangement. Each slot is labelled with a unique identifier and colour-coded to indicate its occupancy status, where green represents available slots and red represents occupied slots. A summary panel is provided to show the total number of available and occupied spaces, enabling users to quickly assess parking availability at the selected location.

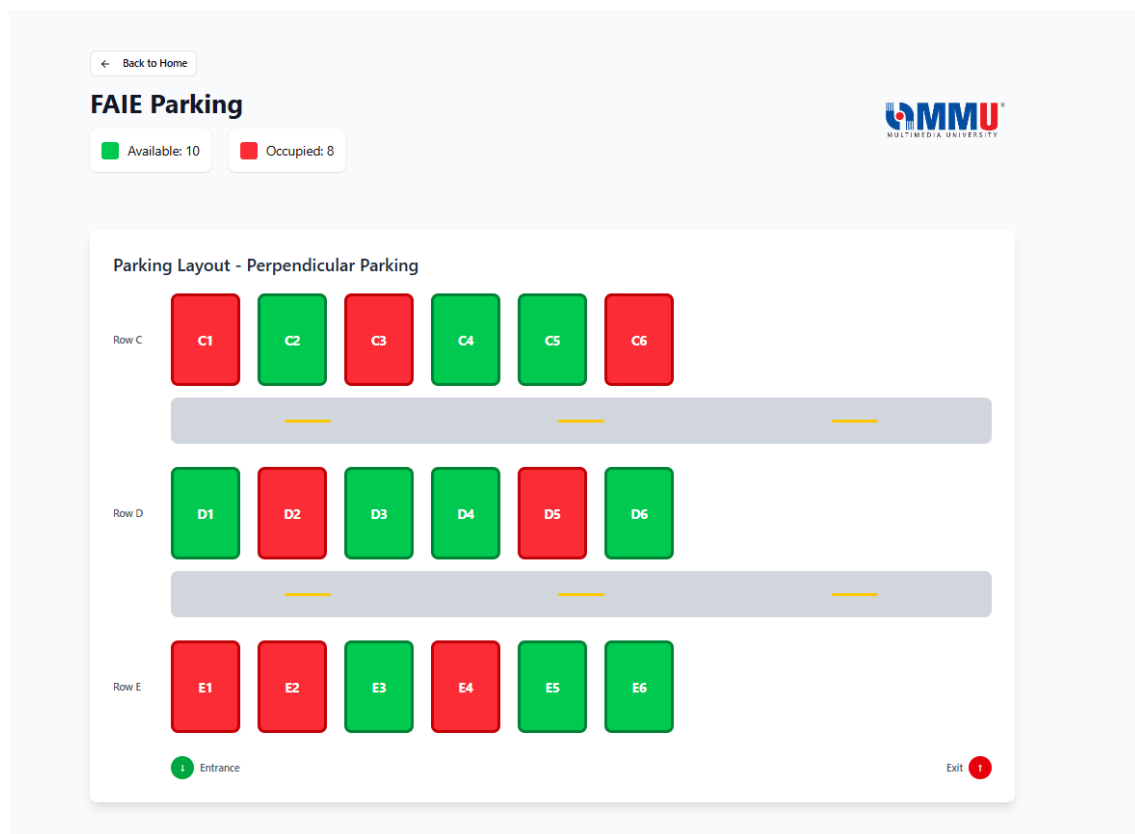


Figure 4.8: FAIE Parking Interface

4.8 Summary

This chapter presented the system design of the proposed parking monitoring system, translating the problem analysis and requirements defined in Chapter 3 into structured design representations. The chapter began with a Rich Picture Diagram to provide a high-level understanding of the parking problem environment and the proposed solution within the MMU campus context. This conceptual view helped connect real-world parking challenges with the technical approach adopted in this project.

Subsequently, a Use Case Diagram was developed to illustrate how users interact with the system. The interaction was intentionally kept minimal and view-based, reflecting the project scope and avoiding unnecessary application-level complexity. The Activity Diagram was then used to describe the end-to-end workflow of the system. Showing how video feeds are processed by the detection model and how parking availability information is updated and displayed on the dashboard.

The Class Diagram defined the core data entities and their relationships, supporting the implementation of multiple parking locations while maintaining a lightweight design. The Sequence Diagram further detailed the dynamic interactions between system components including location selection, frame processing, occupancy detection and dashboard updates.

Finally, the Interface Design section presented the dashboard layouts for both FCI and FAIE parking areas. Demonstrating how real parking geometries are visually represented and how users can intuitively interpret parking availability. This chapter established a coherent and consistent system design that aligns with the project objectives and requirements. The designs presented serve as a solid foundation for the implementation and evaluation phases discussed in the subsequent chapter.

Chapter 5: Implementation Plan

5.1 System Implementation Overview

This chapter outlines how the parking monitoring system which was designed in Chapter 4 was implemented. This chapter aims to explain how the overall components proposed in the system are realized and how they interact to work together in processing the parking video information to provide parking availability information to users.

The implemented system concentrates on three key areas, which include:

- a. Video data collection
- b. Parking occupancy detection
- c. dashboard implementation & visualization

Video images of specific parking spots are processed to get image frames that are analyzed by a parking occupancy detection model to display the status of a given parking slot. Detection results are then displayed in order to update the dashboard interface; users are then able to see parking layout, slot availability and summary of various parking locations.

This chapter is structured in the following way. In section 5.2, the data acquisition of a video and frames to analyze it is detailed. In section 5.3, the parking occupancy detection model is presented and how it is used to identify slot status. The mapping of the results of detection to parking slots and their re-updating in the system is discussed in Section 5.4. Section 5.5 describes the application of the dashboard interface, layout visualization, and selection of location. Lastly, Section 5.6 describes the tools and technologies of the system implementation.

5.2 Video Data Collection

In this section, one will explain the process of recording video data and making it ready to be processed in the parking monitoring system. **The main source of input data in this project is pre-recorded video files for the parking occupancy detection.** Recording the videos allows the system to be tested in a controlled way and to guarantee the same level of input of the system and a consistent analysis of the videos in the different parking areas.

The use of pre-recorded videos was chosen due to a number of practical reasons. To start with, time limits are related to the time frame of the project and it is not possible to implement and test a complete real-time camera system.

Second, the lack of resources limits access to the required hardware facilities, e.g. fixed cameras and networking devices, to facilitate constant real-time video delivery.

Lastly, the chosen parking areas on the MMU campus currently cannot be directly integrated with live parking footage no matter how hard people might want to do this, as there is no publicly accessible real-time video feed of the selected parking lots.

Regardless of these drawbacks, videotaped videos can still offer realistic parking situations and can be used to properly test the functionality of the system. The targeted parking lots, FCI and FAIE will be videotaped in order to capture the common parking situations using either a flying drone or phone camera. The recordings will reflect actual parking situations, such as different degrees of traffic, car flow and fluctuation of parking slot occupancy. The system works with each video file separately and is associated with a particular parking point.

The system will retrieve the recorded video files one at a time and get separate frames at a specified time frame during implementation. Frame extraction allows the system to examine the parking occupancy frame by frame with the help of computer vision. The method is less complex to process but still

provides enough progressive resolution to predict changes in the parking slot status.

The frames are then extracted and sent into the parking occupancy detection module where they are further processed. The system is designed using a modular structure that can be extended in the future, e.g., by replacing recorded videos with live camera feeds, by separating the actions of video collection and frame extraction and the detection step. Such an implementation approach will guarantee that the system is capable of reliably processing the parking data and at the same time is in accordance with the functional requirements defined.

5.3 Parking Occupancy Detection Model

This section outlines the parking occupancy detect system that is deployed in the system. This component aims at analyzing video frames retrieved by use of recorded parking footage and establishing the occupancy rate of the individual parking slots. The computer vision-based car park slot detection system is implemented as its design is quite robust and able to identify parking lots in changeable conditions.

The system uses a vehicle detection model that is based on YOLO as the main detection system. The choice of YOLO (You Only Look Once) is based on its good performance in object detection, especially in vehicle detection in the real world. The model can handle image frames in an efficient way and retain the credibility of detection accuracy and it is the appropriate model to use in the frame-by-frame analysis of parking video data. The implementation of the system uses **the most dependable and up-to-date version of the YOLO model that existed during the time of development**, which enables the ability to choose the best version during the implementation.

The YOLO model is then used to identify vehicles in the parking space on a per-frame basis. Bounding boxes, class labels and scores of confidence of detected vehicles are the output of the detection model. The occupancy of parking slots in the system is determined by using these detection results.

The system uses vehicle detection to map vehicles to pre-defined parking slots areas using the physical structure of each parking site to determine the occupancy of individual parking slots. The parking slot areas are delimited based on the realistic layout of slots in FCI and FAIE and support of angled parking patterns. The parking slot is considered as occupied when the vehicle identified occupies a large portion of the slot area. In case a vehicle overlap is not detected the slot is considered empty. This mapping method allows determining occupancy at the slot level without having to detect a different occupancy model per parking layout.

The system design can be configured to optionally combine a second classification method in cases where vehicle detection performance is subject to influence by factors like occlusion, changes in lighting or by perspective distortion. In the event that this is needed in the implementation, a lightweight convolutional neural network (CNN) can be utilized to categorize each parking slot as either occupied or empty by cropped slot regions. The secondary model is considered to be a backup mechanism to make it more robust in tricky situations instead of having to replace the default YOLO-based detection.

In general, the chosen detection strategy yields a decent trade-off between accuracy, efficiency, as well as implementation feasibility. Through a combination of vehicle detection by the means of the YOLO algorithm and mapping the parking slots, the system successfully identifies the parking occupancy in various parking schemes and is still open-ended to future upgrades.

5.4 Parking Slot Mapping and Status Update

This section explains how vehicle detection results are mapped to individual parking slots and how the occupancy status of each slot is updated within the system. This process plays a critical role in translating raw detection outputs into meaningful parking availability information that can be displayed on the dashboard.

5.4.1 Parking Slot Region Definition

Each car park slot at the areas being monitored (FCI and FAIE) is taken by a predefined area that is taken as the physical location of the car park slot in the parking premises. These areas are created according to the real parking scheme and are fixed during the system execution. **In perpendicular parking layouts like FAIE, slot areas are represented by a rectangular area, whereas in angled parking layouts like FCI, slot areas are represented by polygons** which are more appropriate to represent the orientation of parking slots.

Parking slots are given their own identifiers (e.g. A1, B2) and linked with their appropriate area. This preset mapping is used to guarantee that vehicles identified have the opportunity of being precisely tied to particular parking spots despite variations in the arrangement or direction of parking.

5.4.2 Mapping Detection Results to Parking Slots

Once a vehicle detection is done on every frame, the system is fed a set of bounding boxes which are detected vehicles. The system uses the spatial relationship of the detected vehicle bounding boxes and the predefined parking slot regions to obtain occupancy of parking slots.

The overlaps of the vehicle bounding box and the parking slots regions are calculated against each vehicle they represent, which are detected. The geometric intersection method, including intersection-over-area or intersection-

over-union (IoU), is used to measure this overlap. **In the event that the area of a parked vehicle and a parking slot area intersect more than a pre-determined threshold, then the area is declared occupied.** In case of lack of enough overlap, the parking slot is labelled as empty.

This method enables the system to accommodate both perpendicular and angled parking designs with a single detection and mapping plan without having to create separate detection models in various parking designs.

5.4.3 Parking Slot Status Update Algorithm

The algorithm to be used to update the parking slot occupancy status of each processed frame can be summarized as follows:

- a. Get the following video frame in the video feed being recorded.
- b. Use the vehicle detection model based on YOLO to the frame and extract vehicle bounding boxes.
- c. In every defined area of parking slots:
 - i. Calculate the intersection between the slot area and the vehicle bounding boxes that are found.
 - ii. In case the overlap is greater than the specified threshold, cross the slot.
 - iii. Otherwise, cross the slot as vacant.
- d. Provision of the occupancy status of every parking slot depending on the evaluation outcomes.
- e. Re-calculate the number of occupied and available parking slots of the chosen location.
- f. Send revised slot status to the dashboard to be visualized.

This is done with each successive frame so that the system can update its information on parking availability as the vehicle poses continually.

5.4.4 Status Synchronization with Dashboard

After statuses of parking slots are updated, the system updates the most recent status on occupancy with the dashboard interface. The visual updating of every parking slot with color indicators reflects its status and summary counts are recalculated to give an overview of parking availability at the chosen location. This will guarantee that the users get updated and consistent information that is a true representation of the observed parking conditions.

5.5 Dashboard Implementation

This part explains how the dashboard part of the parking monitoring system is implemented. The major user interface is presented by the dashboard where the information about the availability of parking places is available visually through the occupancy detection process. The dashboard design and implementation are informed by the user requirements outlined in Chapter 3 and interface designs outlined in Chapter 4.

Dashboard is designed with the location-based viewing format in which the user can switch among the areas monitored by pressing tabs or buttons. During the implementation, there are two parking points served: FCI and FAIE. In the case of an area being chosen, the dashboard will retrieve and show the parking plan and occupancy data of that area. This design allows easy navigation and has a clean and uncluttered interface.

The dashboard displays parking slots in their physical location in the chosen parking space visually. Each slot has a special identifier and it is represented as a polygonal or a rectangular structure, which depends on the parking organization. Color indicators are used to communicate slot status with a green slot meaning an empty slot and red slot meaning occupied slot. With this visual representation, the users are able to interpret the availability of parking in a short time without the need to analyze it.

Along with slot-level visualization, the dashboard also offers a summary panel that is used to show the number of parking slots available and occupied at the location of choice. These summary values are dynamically calculated with the latest result of occupancy detection. The dashboard provides detailed slots information as well as a high-level overview, which allows making a quick decision as well as taking a closer look at parking conditions.

The implementation of the dashboard will be responsive to changes provided by the parking occupancy detection module. Each time new frame analysis results are available, slot statuses and summary counts are updated to show the latest parking conditions. This makes the detection output and information that is delivered to the users consistent. All in all, the implementation of the dashboard converts the technical outcomes of detection into a simple and easy-to-read visualization, making it possible to monitor the availability of parking places at various places of parking.

5.6 Tools and Technologies

In this section, the tools and technologies that were employed in the implementation process of the parking monitoring system are outlined. The tools selected were done depending on the ability to perform tasks of computer vision, the ability to integrate easily and the availability of the tools within the project limits.

5.6.1 Programming Language

- Python
 - The main programming language to implement the system.
 - Chosen because it offers a wide range of support of computer vision and machine learning libraries.
 - Gives video processing, model integration, and data processing flexibility.

5.6.2 Video Processing

- OpenCV
 - Used to work with recorded video files.
 - Responsible for:
 - Reading video inputs
 - Drawing frames at specified rates.
 - Adjusting simple image processing tasks.
 - The system also allows analysis between frames which are necessary in parking occupancy detection.

5.6.3 Parking Occupancy Detection

- Object Detection Model based on YOLO (Ultralytics family).
 - Used to identify cars in the extracted frames of video.
 - Chosen due to:
 - High object detection performance under real world situations.
 - Reasonable balance between detection and computing efficiency.
 - During implementation, the particular version of YOLO is chosen in terms of reliability and compatibility with the system.

5.6.4 Mapping Parking Slots and Determining Statuses.

- Python-based Geometric Operations.
 - Applicable to calculate spatial relationships between:
 - Bounding boxes of identified vehicles
 - Pre-defined areas of parking slots.
 - Facilities correct categorization of parking slots as:
 - Occupied

- Empty
- Profiles perpendicular (FAIE) and angled (FCI) parking.

5.6.5 Dashboard Visualization

- Lightweight Visualization Approach.
 - Used to render:
 - Parking layouts
 - Slot representations
 - Occupancy summary data.
 - Concentrates on clarity, usability and ease of interpretation.
 - Makes the interface user-friendly and offers good visual feedback.

In general, the chosen tools and technologies can offer an efficient and viable development environment to implement the parking monitoring system. They are able to support the necessary video processing, occupancy detection and visualization tasks and are able to be extended and enhanced in future.

5.7 Summary

This chapter explained how the parking monitoring system will be implemented as per the system design in Chapter 4 was done. This chapter was dedicated to describe the understanding of the system components and their interaction to process the parking video data and display the parking availability information to the users.

Some of the key implementation issues in this chapter are:

5.7.1 Video Data Collection

- The system will be fed with videotaped files.
- Videos on FCI and FAIE parking will be separately processed.

- Video recordings will be taken and frames will be cut out to allow frame-by-frame analysis.

5.7.2 Parking Occupancy Detection.

- Vehicles will be detected in the individual frame by using a YOLO based vehicle detection model.
- Vehicles detected will be analyzed in order to identify the occupancy of parking slots.
- The system design provides the ability to choose the most reliable and documented YOLO version at the implementation stage.

5.7.3 Status Update and Parking Slot Mapping.

- The parking lots will be modeled with pre-coded areas regarding the real parking plans.
- Cars parked will be identified and geometrical overlaps is going to be calculated to find parking spots.
- The slot status became dynamically updated as either occupied or empty corresponding to each processed frame.

5.7.4 Dashboard Implementation

- A dashboard system is adopted to visualize parking spaces.
- There is an option of switching between FCI and FAIE parking places.
- There is clear parking layout, color coded slot status and summary counts.

5.7.5 Tools and Technologies

- Python is the primary programming language.
- OpenCV will be used to process video and extract video frames.
- Vehicle detection will be implemented by a YOLO based detection model.
- The dashboard interface will be rendered using lightweight visualization techniques.

On the whole, this chapter showed how the proposed system was gathered in a systematic and realistic way, which will guarantee compliance with the functional requirements and system design. The implementation, as well as presented in this chapter, leaves a good basis to assess performance and effectiveness of the system as addressed in the following chapter.

Chapter 6: Conclusion

6.1 Project Summary

The proposed project seeks to deal with the challenge of congestion around the parking areas and the inconvenience of finding a parking space in the MMU campus. As it has been found in the course of preliminary research, there is a common situation when the user wastes time, feels frustrated, and even uncertain during the process of seeking the parking, especially during the busy time. The lack of information on the availability of clear and timely parking places is the primary cause of these problems.

In order to get a better comprehension of the issue, the survey and observational study that was undertaken by this project centered on the parking places that are widely used in the campus, which are FCI and FAIE. The survey results and observations pointed out the common parking issues, which consist of significant congestion during the midday times and ineffective traffic flow within the parking spaces. These observations formed the basis of the suggestion of a parking monitoring system using the camera and applying computer vision techniques.

The suggested system will be based on video detection of vehicles to deduce the occupancy of parking slot and displaying the availability of parking slot in a visual interface. Even though this system is yet to be implemented, this project has already laid out an extensive analysis, design and implementation plan, which will be used to guide the development in the upcoming phase.

6.2 Achievement of Project Objectives

This project objectives were mainly aimed towards analyzing the problem, designing the system and planning the implementation as opposed to deploying the system for FYP 1. The project goals are met at this point at conceptual and design stages. Particularly, the project has:

- a. Determined the major parking-related challenges using surveys and firsthand observations
- b. Examined user expectations and usage behaviors of parking in the selected campus locations namely FCI and FAIE.
- c. An option was to implement a camera-based parking monitoring system to deal with the problems identified.
- d. Clear functional and non-functional requirements that are defined according to the actual user feedback.
- e. Developed structured system drawings such as use case, activity, class, sequence and interface drawings.
- f. Came up with a systematic system implementation strategy, which included data acquisition, vehicle detection, mapping parking slots and dashboard visualization.

Although the system is yet to be implemented and tested, the system analysis, design and planning objectives have been achieved. These will help in the seamless passing of the implementation phase in the subsequent phase of the project.

6.3 Key Outcomes and Current Limitations

The most important deliverable of the project is the creation of a systematic and well-grounded plan of a parking surveillance system in the MMU campus setting. The project has gained an explicit overview of the current parking issues, expectations of its users, and high-demand parking spots through survey analysis and direct observation. These were the findings on the basis of which system requirements were determined and a system design and implementation plan were developed.

A number of restrictions are present at the current stage, and they are in line with the restrictions mentioned in the previous section, Section 5.2 (Video Data Acquisition).

First, the utilization of video data records as opposed to the live video camera feed has been discovered to be a design requirement because of the time restrictions that are linked with the timeframe of the project and constraints on resources, specifically the unavailability of specific camera equipment and other infrastructure. Moreover, the real-time video feeds of the chosen parking places in the MMU campus are not publicly available, which further limits the viability of the real-time implementation in this stage.

Lastly, it is purposefully narrowed in specifically to the chosen parking points that include FCI and FAIE, due to the survey results that show that these areas are problematic and in demand. This is a restricted scope necessary to keep the system design within the project scope and yet solve a real and valid issue.

These limitations do not define as weaknesses of the project but rather, they are workable restrictions that define the scope and the direction of the work. They form a rational rationale of the implemented methodology and they create a realistic base of the system development and testing in the following stage of the project.

6.4 Future Work and Next Phase (FYP2)

The second stage of the project (FYP2) is going to be devoted to the implementation and testing of the offered computer vision system. A few major functions will be involved in this phase.

To ensure the input, the video data will be recorded in specific parking places and represented as an input to the implementation of the systems. The vehicle detection model shall then be incorporated into the system via a YOLO-based vehicle detection model to conduct frame-by-frame analysis of vehicle detection to determine the occupancy of car parks.

The system will then apply parking slot mapping methods to get occupancy of slots according to the detected vehicles. In the dashboard interface, the visualization of parking layouts, color-coded slot availability and summary information about the various parking locations will be developed. Performance evaluation will also be done to meet the detection accuracy, system responsiveness and robustness under different lighting and traffic conditions.

Further improvements can also be made in the form of live camera feeds, extension to other parking areas, inclusion of other new features like historical parking analysis, or alert systems. They will make the proposed system more practical and scalable.

6.5 Concluding Remarks

To summarize this project, it has provided a strong base upon which a computer-vision based car park slot detection system can be developed in MMU campus. The problem analysis, requirement definition, system design and implementation plan have helped the project to have a clear and feasible direction towards solving parking challenges in the campus set up.

Even though no system implementation has been undertaken during this stage, the efforts undertaken are a comprehensive road map of further development. The results, layouts and layouts found in this report will become a useful guide to the implementation and assessment phases that will be done in FYP2.

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Appendix A: Gantt Chart

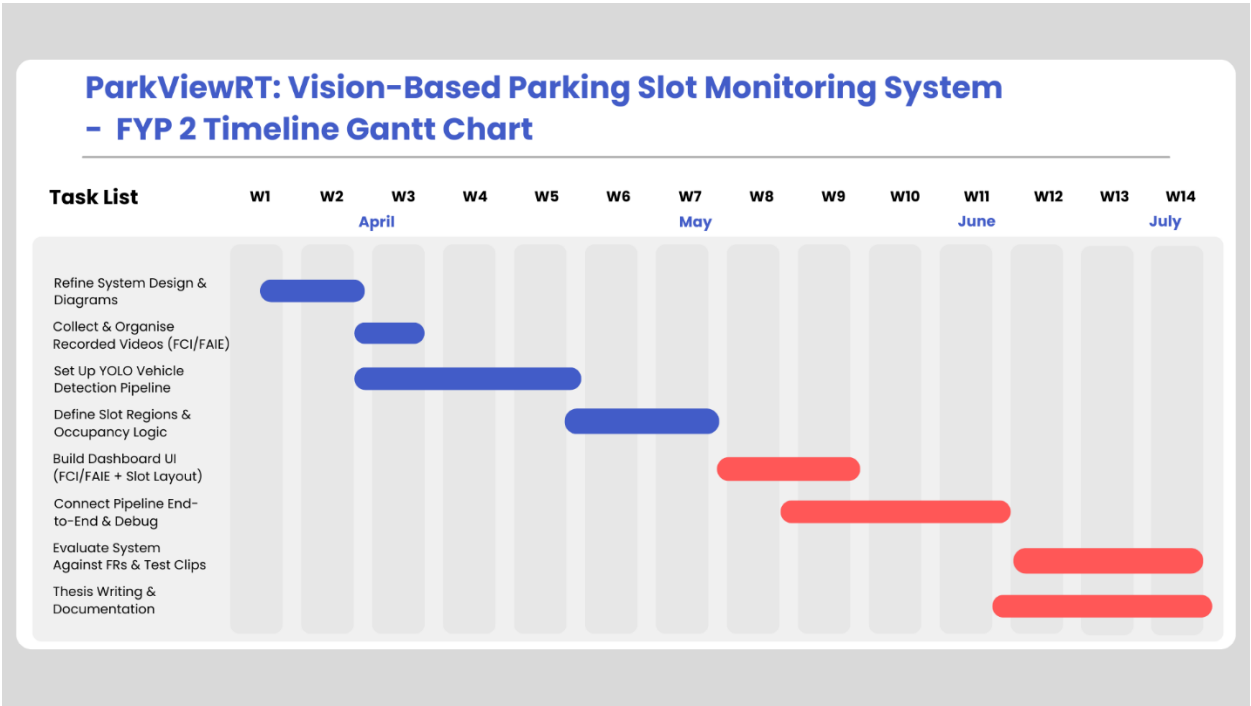


Figure: Gantt Chart for FYP2

Appendix B: FYP I Meeting Logs



CPT6314 Final Year Project (FYP) 1 Meeting Log

Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 19/11/2025	Meeting No.: 1
Meeting Mode: Face-to-Face	
Project ID: 833	Project Type: Application-based
Project Title : ParkViewRT: Vision-Based Parking Slot Monitoring System	
Student ID : 1221305612	Student Name: Yaser E H Abulaban
Student Programme and Specialisation: Computer Science (Data Science)	
Supervisor Name: Mr. Md. Zubair Hassan Tarif	Co-Supervisor Name:(if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name:(if applicable)

1. WORK DONE

Details (in point form):

- Kick off meeting was held to further explain the project scope and path to start
- Parking was identified along side with recording mechanism

2. WORK TO BE DONE**Details (in point form):**

- Identifying research papers related to the project's scope
- Selecting 10 research papers as a start to get familiarized with the tools used related to Parking systems
- Hold the second meeting with Mr. Zubair to identify further steps to be taken and discuss the findings from these 10 papers

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

- So far, no issues have been encountered yet

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

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Multimedia University, Cyberjaya 63100, Selangor, Malaysia



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Supervisor's Signature

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Student's Signature

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Co-Supervisor's Signature

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Company Supervisor's Signature

(if applicable)**(if applicable)**

IMPORTANT NOTES TO STUDENTS:

- 1) Items 1 – 3 are to be completed by the students prior to the meeting. Item 4 is to be completed by the supervisor / co-supervisor / company supervisor.
- 2) Student has to upload the soft copies of the meeting logs in eBwise and also attach them along with interim (FYP1) report.
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- 3) Log sheets provide the basis for evaluating the General Effort (Project Management, Attitude, and Technical Competency) of the student, by the supervisor and also for checking the attendance requirement of the student, by the FYP Committee.
This also provide the student with feedback from the supervisor / co-supervisor / company supervisor on the tasks done and provide the plan for the upcoming tasks. This can provide the motivation for the student to give consistent and efficient effort throughout the period of FYP.
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CPT6314 Final Year Project (FYP) 1 Meeting Log

Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 19/11/2025	Meeting No.: 2
Meeting Mode: Online	
Project ID: 833	Project Type: Application-based
Project Title : ParkViewRT: Vision-Based Parking Slot Monitoring System	
Student ID : 1221305612	Student Name: Yaser E H Abulaban
Student Programme and Specialisation: Computer Science (Data Science)	
Supervisor Name: Mr. Md. Zubair Hassan Tarif	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

Details (in point form):

- Collected 10 research papers related to my project topic to gain further insights on different solutions implemented towards solving the issue of parking
- Discussed couple of approaches to be taken to decide on modules to be used in this project
- Identified the type of datasets to gather for module training purposes

2. WORK TO BE DONE

Details (in point form):

- Start summarizing the main ideas and problems discussed in the 10 research papers that were collected as discussed in the first meeting.
- Start building Chapter II in the report (Literature Review).
- Identify datasets to be used for module training.
- Hold a third meeting with Mr. Zubair to discuss the progress of Chapter II in the report and datasets

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

- So far, no issues have been encountered yet

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



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Co-Supervisor's Signature

(if applicable)



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Student's Signature

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CPT6314 Final Year Project (FYP) 1 Meeting Log

Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 08/12/2025	Meeting No.: 3
Meeting Mode: Online	
Project ID: 833	Project Type: Application-based
Project Title : ParkViewRT: Vision-Based Parking Slot Monitoring System	
Student ID : 1221305612	Student Name: Yaser E H Abulaban
Student Programme and Specialisation: Computer Science (Data Science)	
Supervisor Name: Mr. Md. Zubair Hassan Tarif	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

Details (in point form):

- Presented 1st draft of Chapter II
- Presented the collected research papers
- Presented the collected Datasets
- Initial selection of models to be used based on the data collected
- Discussed ideas for chapter I & III

2. WORK TO BE DONE

Details (in point form):

- Collect 10 to 15 more research papers and expand chapter II
- Deep-dive into used frame works/models used for the project for Chapter III
- Start working on Chapter I & III

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

- So far, no issues have been encountered yet

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



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Student's Signature

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Co-Supervisor's Signature

(if applicable)

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Company Supervisor's Signature

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CPT6314 Final Year Project (FYP) 1 Meeting Log

Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 31/12/2025	Meeting No.: 4
Meeting Mode: Online	
Project ID: 833	Project Type: Application-based
Project Title : ParkViewRT: Vision-Based Parking Slot Monitoring System	
Student ID : 1221305612	Student Name: Yaser E H Abulaban
Student Programme and Specialisation: Computer Science (Data Science)	
Supervisor Name: Mr. Md. Zubair Hassan Tarif	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

Details (in point form):

- Presented Chapters I & II
- Discussed the current progress of Chapter III
- Discussed some potential enhancements/adjustments to Chapters I & II

2. WORK TO BE DONE**Details (in point form):**

- Revise the discussed enhancements/adjustments to Chapters I & II
- Designing a survey for Chapter III and to target between 60 – 100 responses
- Discussing the survey before publishing it
- After getting the results of the survey, analyze the results and finish Chapter III

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

- So far, no issues have been encountered yet

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

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CPT6314 Final Year Project (FYP) 1 Meeting Log

Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 08/01/2026	Meeting No.: 5
Meeting Mode: Online	
Project ID: 833	Project Type: Application-based
Project Title : ParkViewRT: Vision-Based Parking Slot Monitoring System	
Student ID : 1221305612	Student Name: Yaser E H Abulaban
Student Programme and Specialisation: Computer Science (Data Science)	
Supervisor Name: Mr. Md. Zubair Hassan Tarif	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

Details (in point form):

- Presented Chapters III
- Presented the survey
- Discussed the current progress of Chapter III
- Discussed some potential enhancements/adjustments to Chapter III & the survey
- Discussed the approach to be taken for the Observation study

2. WORK TO BE DONE

Details (in point form):

- Revise the discussed enhancements/adjustments to Chapter III & the survey
- Distributing the survey among MMU students/staff and aim to get 60 – 100 responses
- After getting the results of the survey, analyze the results and finish Chapter III
-

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

- Sample size is a bit high
- As a solution I need to mass send to all the groups I am in that is for MMU (study groups/lecture groups/ event groups etc.) and make sure to email students and staff to do the survey

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)



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CPT6314 Final Year Project (FYP) 1 Meeting Log

Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 22/01/2025	Meeting No.: 6
Meeting Mode: Online	
Project ID: 833	Project Type: Application-based
Project Title : ParkViewRT: Vision-Based Parking Slot Monitoring System	
Student ID : 1221305612	Student Name: Yaser E H Abulaban
Student Programme and Specialisation: Computer Science (Data Science)	
Supervisor Name: Mr. Md. Zubair Hassan Tarif	Co-Supervisor Name: (if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name: (if applicable)

1. WORK DONE

Details (in point form):

- Presented the findings of the survey and observation study
- Discussed some adjustments to be made for functional requirements in chapter III
- Presented Chapter IV

2. WORK TO BE DONE**Details (in point form):**

- Expand more on Functional Requirements to have minimum 10, each existing FR should be broken down to small tasks not big ones that have more than one requirement in it
- Chapter IV looks nice – no adjustments needed
- We need to suggest places on where to mount the cameras in MMU (there's a mounted camera between FCI and library Parking – I need to take a picture of it)
- Finalize the adjustments for chapter III
- Start working on Chapter V (implementation Plan) and Chapter VI (Conclusion)

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

- In order to suggest places to mount the cameras, I need to go inside some classes/lecture offices to check the possibility of it
- As a solution I need to request permission from the faculty (FCI & FAIE) to do that and I might need to ask my Lecturers for permissions to enter their offices.

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

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CPT6314 Final Year Project (FYP) 1 Meeting Log

Trimester OCT / NOV 2025 (Trimester ID:2530)

Meeting Date: 05/02/2026	Meeting No.: 7
Meeting Mode: Online	
Project ID: 833	Project Type: Application-based
Project Title : ParkViewRT: Vision-Based Parking Slot Monitoring System	
Student ID : 1221305612	Student Name: Yaser E H Abulaban
Student Programme and Specialisation: Computer Science (Data Science)	
Supervisor Name: Mr. Md. Zubair Hassan Tarif	Co-Supervisor Name:(if applicable)
Collaborating Company: (if applicable)	Company Supervisor Name:(if applicable)

1. WORK DONE

Details (in point form):

- Presented Chapters V & VI
- Presented the suggested adjustments for Chapter III that were discussed last meeting (Meeting 6)
- Discussed the Gantt chart
- The report is 95% done, few minor things to complete

2. WORK TO BE DONE**Details (in point form):**

- Build the Gantt chart that should reflect timelines for FYP2
- Fill in Abstract, Acknowledgement and other small sections
- Start building the slides for the presentation

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

- So far, no issues have been encountered yet

4. COMMENTS (Supervisor / Co-Supervisor / Company Supervisor)

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Appendix C: Turnitin Similarity Index Page

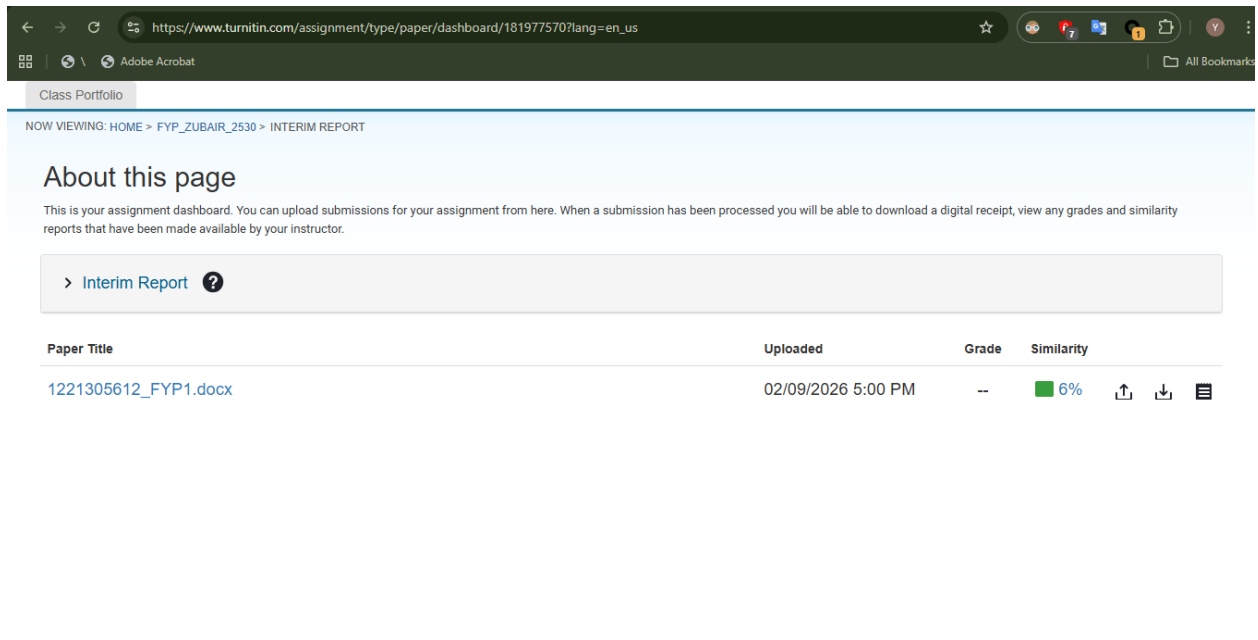


Figure: Tunitin Similarity %